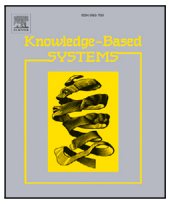
Knowledge-Based Systems 343 (2026) 116034



Contents lists available at ScienceDirect

Knowledge-Based Systems

journal homepage: www.elsevier.com/locate/knosys





Agentic LLM for anonymizing healthcare data with contextual awareness

Rana Azzam [a,*], Ahmad Musamih [a], Senay Gebreab [b], Khaled Salah [b], Mohammed Omar [a]

[a] Department of Management Science and Engineering, Khalifa University, Abu Dhabi, United Arab Emirates
[b] Department of Computer and Information Engineering, Khalifa University, Abu Dhabi, United Arab Emirates

ARTICLE INFO

Keywords:
Agentic AI
Large language models (LLMs)
Anonymization
Healthcare data
Privacy

ABSTRACT

Large language models (LLMs) are transforming healthcare applications by enhancing data analysis, yet their adoption remains constrained by stringent privacy requirements that limit access to sensitive medical data. Anonymization offers a pathway to address this challenge; however, existing techniques often lack contextual understanding and exhibit low accuracy, compromising either patient privacy or the utility of clinical content. In this paper, we propose an end-to-end, modular agentic LLM system for processing sensitive healthcare data with contextual awareness. The system is orchestrated by a central agent coordinating specialized components for structured data retrieval, clinical narrative generation, anonymization, public LLM querying, and secure deanonymization. Locally hosted LLMs handle all privacy-sensitive steps, including context generation and anonymization, while public LLMs are used exclusively for reasoning on pre-anonymized inputs.

We evaluate our system on a synthetic clinical dataset and benchmark it against five state-of-the-art named entity recognition (NER) techniques. Our approach achieves high precision and a recall of 93.6%, significantly outperforming baselines such as spaCy (33% precision, 89% recall) and Presidio (41% precision, 90% recall). Additional evaluation on real-world clinical notes from the MIMIC-III dataset demonstrates strong generalization to unstructured narratives, achieving a clean transformation rate of 98.6%. We further evaluate medical richness, showing that anonymized outputs retain clinically relevant information and semantic structure, preserving downstream utility. Adversarial re-identification experiments confirm that no true identifiers can be reconstructed, highlighting the framework's effectiveness in balancing privacy, robustness, and clinical usefulness.

1. Introduction

The widespread adoption of electronic medical records (EMRs) over the past decade has laid the foundation for the development of smart healthcare systems, particularly those leveraging data-driven large language models (LLMs) [1]. The availability, seamless accessibility, retrieval immediacy, and variety of clinical data have accelerated advancements in this field, resulting in substantial and promising outcomes. Systems powered by LLMs represent a transformative approach in the healthcare industry [2]. These systems leverage the vast amounts of medical knowledge embedded in EMRs and other clinical data sources to assist healthcare professionals in various tasks, such as making more accurate, evidence-based decisions. LLMs, trained on diverse medical literature and patient records, can process and interpret complex data, offering real-time, context-specific insights during clinical workflows. Healthcare systems based on LLMs enhance decision-making by providing professionals with tailored recommendations based on patient data, improving diagnosis and treatment plans [3]. These systems offer real-time assistance, enabling quick decisions in critical situations. They are continually updated with the latest medical knowledge, which ensures professionals have access to current information. Additionally, LLM-based healthcare systems help reduce human error by identifying patterns and risks that may be overlooked by human professionals, which ultimately enhances patient safety [4]. Despite these benefits, the sensitive nature of clinical data poses a serious barrier to deploying LLMs in practice. In 2023 alone, over 168 million healthcare records were compromised in 725 reported data breaches, with nearly 80% attributed to hacking and IT incidents [5]. These alarming figures underscore the growing vulnerability of healthcare systems and highlight the urgent need for robust privacy protections when integrating LLMs into clinical workflows.

One of the primary concerns when integrating public LLMs into healthcare workflows is data privacy. The use of sensitive patient data introduces significant risks of unauthorized access and regulatory violations, particularly under frameworks such as the Health Insurance

* Correspondence to: Khalifa University, Abu Dhabi, United Arab Emirates.
E-mail addresses: rana.azzam@ku.ac.ae (R. Azzam), ahmad.musamih@ku.ac.ae (A. Musamih), senay.gebreab@ku.ac.ae (S. Gebreab), khaled.salah@ku.ac.ae (K. Salah), mohammed.omar@ku.ac.ae (M. Omar).

https://doi.org/10.1016/j.knosys.2026.116034
Received 17 July 2025; Received in revised form 11 April 2026; Accepted 15 April 2026
Available online 18 April 2026
0950-7051/© 2026 Published by Elsevier B.V. This is an open access article under the CC BY license (http://creativecommons.org/licenses/by/4.0/).

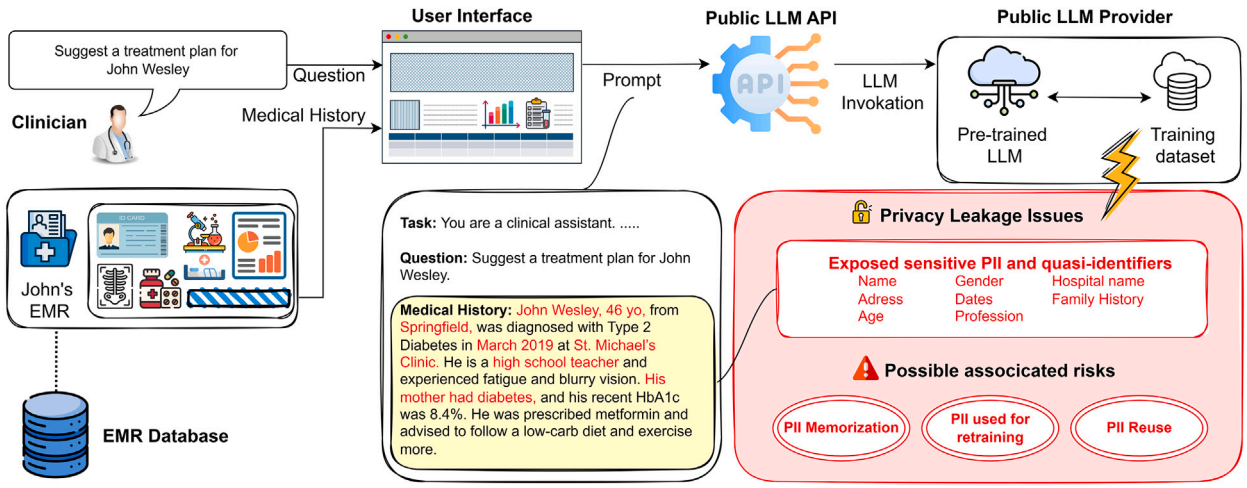


Fig. 1. Conceptual diagram illustrating the privacy issues of patient's healthcare sensitive data when using LLMs.

Portability and Accountability Act (HIPAA) [6] and the General Data Protection Regulation (GDPR) [7], as illustrated in Fig. 1. As a result, anonymization has become an essential pre-processing step to protect sensitive patient data and enable the use of powerful LLMs.

While there are established anonymization techniques such rule-based systems and machine learning-based names entity recognition approaches, these methods struggle to find the balance between data privacy and clinical utility. Specifically, many approaches lack robust contextual reasoning, leading either to incomplete removal of sensitive identifiers or excessive redaction that degrades medical interpretability. This trade-off between privacy preservation and data utility remains a central challenge in clinical text processing [8,9].

Given these challenges, more advanced healthcare systems must be carefully designed to realize the full potential of LLMs. LLMs can be deployed in both public and local domains. Locally hosted LLMs operate within secure, isolated systems and are not accessible by external entities, making them well-suited for processing sensitive data while upholding strict privacy guarantees [10]. However, due to limitations in training data scope and computational resources, local LLMs often lack the breadth and generalization capabilities of publicly hosted LLMs [11], which are particularly valuable in healthcare systems requiring robust clinical reasoning across diverse patient populations. As a result, their outputs may require additional validation and refinement to ensure they meet domain-specific expectations and task requirements. When validation and refinement are systematically combined, they enable more reliable and controllable LLM-powered systems.

Such systems cannot rely on a single LLM alone; instead, a structured process involving planning, reasoning, action, and refinement is essential. This highlights the need for agentic architectures [12], which provide a modular and interpretable framework for orchestrating complex workflows. Agentic LLM architectures enable more controllable, auditable, and reliable pipelines. They decompose tasks into specialized, coordinated sub-tasks. Their ability to integrate diverse resources — such as tools, databases, and external knowledge sources — is particularly valuable in healthcare settings, where domain-specific context is crucial for accurate, evidence-based decision-making.

In healthcare settings, agentic architectures enhance clinical decision-making by systematically analyzing relevant medical data and enabling context-aware recommendations. These architectures support iterative refinement, allowing system outputs to be continuously validated against predefined criteria, including privacy and regulatory requirements. Such structured validation mechanisms help ensure that sensitive patient data remains protected while preserving clinical utility. By incorporating these cycles of refinement and validation, agentic LLM frameworks can deliver high-quality, secure, and trustworthy solutions in critical decision support applications. Despite these advances, there is a lack of integrated, privacy-preserving healthcare systems that can autonomously operate end-to-end. Most existing solutions treat anonymization and question-answering (QA) as separate processes. This paper addresses that gap by proposing an agentic architecture that embeds anonymization as a core module in the QA pipeline, enabling secure, accurate, and compliant medical question answering using public LLMs. The main contributions of this paper can be summarized as follows:

- We propose an end-to-end agentic LLM framework for privacy-preserving healthcare systems. The approach features orchestrated integration of a modular pipeline comprising agents for structured data querying, context-aware anonymization, public LLM-based querying, and deanonymization.
- We implement reversible anonymization via an SQL-based query–identity mapping mechanism, allowing sensitive data to be anonymized before external inference and securely restored afterward for internal clinical use.
- We demonstrate privacy-preserving context generation using locally hosted LLMs to support public LLM prompting. Our method goes beyond redaction or masking by generating semantically coherent, anonymized narratives that preserve data utility while protecting patient privacy. This improves both the readability and contextual clarity of prompts, leading to more accurate and relevant public LLM responses.
- We evaluate our proposed framework on a synthetic clinical dataset, benchmarking it against five state-of-the-art NER baselines. Our approach exhibits superior anonymization performance, achieving near-perfect precision and 93.6% recall, significantly outperforming the baselines while reducing over-anonymization and preserving clinical utility. We further assess the system on real-world clinical notes from the MIMIC-III dataset, demonstrating strong generalization to unstructured narratives and evaluating medical richness, where results show that anonymized outputs preserve clinically relevant information and maintain high semantic fidelity to the original records.

The remainder of the paper is organized as follows; in Section 2, a survey of the state-of-the-art privacy preservation techniques with a focus on healthcare is provided and potential research gaps are identified. Section 3 presents an overview of the proposed agentic LLM for anonymizing healthcare data with contextual awareness. Details on implementation are provided in Section 4, followed by use cases in Section 5, then evaluations and benchmarking in Section 6. Concluding remarks and future research directions are provided in Section 7.

2. Literature review

This section presents a detailed review of foundational anonymization techniques, with a particular focus on their applications in the healthcare domain. It highlights the progression from classical methods to learning-based techniques, followed by a review of LLM-based approaches. After identifying limitations in existing anonymization strategies, the review concludes by emphasizing the need for agentic LLM-based solutions.

2.1. History of anonymization techniques

Anonymization refers to the process of modifying personal data to prevent the identification of the subjects of this data, either directly or indirectly [13]. The goal is to reduce the risk of re-identification to a level where the data can no longer be linked to a specific person, even when combined with other information. To achieve anonymization, several techniques have been proposed in the literature since the early 2000s. The most foundational among these, which continue to influence the development of privacy-preserving data analysis, include κ-anonymity [14], ℓ-Diversity [15], t-Closeness [16], and Differential Privacy [17]. In a dataset containing sensitive information about individuals, κ-anonymity ensures that each record is indistinguishable from at least $\kappa - 1$ other records with respect to a set of quasi-identifiers. Quasi-identifiers are not considered sensitive information, such as age and gender. However, the robustness of κ-anonymity is limited when the values of sensitive attributes, e.g. disease, within an anonymized group lack diversity or when an attacker possesses background knowledge. To address these limitations, researchers proposed ℓ-diversity, which requires that each anonymized group contains at least ℓ well-represented distinct values for the *sensitive attribute*, thereby reducing the risk of attribute disclosure. Upon further research, it was shown that ℓ-diversity may still be vulnerable to attribute disclosure, particularly in cases where the sensitive attribute values, though distinct, are semantically similar or unevenly distributed. To address this limitation, t-closeness was proposed. This model requires that the distribution of the sensitive attribute within each anonymized group is close to its distribution in the entire dataset, typically measured using a distance metric such as the Earth Mover's Distance [18]. By enforcing this constraint, t-closeness aims to limit the amount of information gain an adversary can obtain about an individual's sensitive attribute. Unlike the aforementioned techniques, Differential Privacy [17] operates on the outputs of database queries rather than modifying the dataset itself. It introduces a carefully calibrated amount of random noise to the query results, based on a defined privacy budget, thereby limiting the amount of information any single individual's data contributes to the output. This approach enhances robustness against adversarial inference attacks, even when attackers possess substantial background knowledge.

2.2. Privacy-preserving approaches in healthcare

Healthcare institutions are required to comply with stringent privacy regulations such as the HIPAA [6] in the United States and GDPR [7] in the European Union. These regulations mandate the protection of PII and protected health information (PHI), aiming to prevent unauthorized access, misuse, or disclosure of sensitive patient data. Compliance with these regulations is critical for legal and ethical reasons and to maintain patient trust in an era of widespread EMRs and data-driven care. Conventional healthcare-tailored privacy-preserving methods can be broadly classified into classical techniques and learning-based approaches. Classical methods include techniques such as κ-anonymity [14], ℓ-diversity [15], t-closeness [16], and differential privacy [17], which provide privacy guarantees through data transformation or noise injection as discussed earlier. Another common category within classical approaches comprises rule-based systems, which rely on predefined patterns, regular expressions, or dictionary lookups to identify and remove PHI. Techniques like redaction, which removes PHI entirely, and masking, which obfuscates values without altering data structure, fall into this rule-based group. While these methods are transparent and simple to implement, they often lack contextual awareness and struggle with unstructured clinical narratives [19], leading to high false positive or false negative rates.

Among the earliest contributions to medical text de-identification is the MITRE Identification Scrubber Toolkit, MIST in short, developed by Neamatullah et al. [20]. MIST combined rule-based and statistical learning methods to identify and remove PII from unstructured clinical text. Conditional random fields (CRFs) were used to label sequences, effectively detecting sensitive entities within the medical narrative.

MIST [20] demonstrated the feasibility of automated de-identification and proved the potential of natural language processing and machine learning to enhance EMR privacy. This has led to more advancements in machine learning and natural language processing approaches, which are more prevalent in modern anonymization research, with most recent methods leveraging supervised or deep learning models to detect and mask sensitive information more flexibly and accurately than rule-based systems.

Machine learning and natural language processing (NLP) techniques have been increasingly adopted to address the shortcomings of rule-based systems. Among pioneering learning based approaches are Conditional Random Fields (CRF), Long Short-Term Memory (LSTM) networks, and Embeddings for Language Models [21] for Named Entity Recognition (NER) [22]. Building on these early learning-based approaches, the advent of transformer architectures pioneered by Vaswani et al. [23] marked a significant leap in the field. The major advantage of transformer-based models is their ability to capture long-range dependencies and contextual clues in text more effectively than traditional learning-based approaches. This enables them to detect obvious PII and indirect personal identifiers as defined by [19] by referring to the context rather than lists of pre-defined terms or look-up tables. The potential of transformer-based anonymization techniques grows further when finetuned for specific tasks/domains. The availability of large medical corpus plays a pivotal role is enhancing the robustness of transformer-based anonymization approaches targeted for healthcare applications, which will be discussed in more detail in the remaining part of this section.

Bidirectional Encoder Representations from Transformers (BERT) [24] is a language representation model that uses bidirectional transformer layers, enabling it to consider context from both preceding and following words in a sentence. BERT can be fine-tuned for a variety of downstream language tasks by adding and tuning one task-specific output layer. Several domain-specific variants have been developed for biomedical applications, including BioBERT [25] and ClinicalBERT [26]. BioBERT [25] exhibits the same structure as BERT, is finetuned using large biomedical datasets (including PubMed abstracts and PubMed Central articles), and is targeted for biomedical question-answering, biomedical named entity recognition, and biomedical relation extraction. Similarly, ClinicalBERT [26] is based on the BERT architecture, but is finetuned using clinical notes from the MIMIC-III clinical notes. ClinicalBERT is targeted for tasks beyond mining structured medical data.

2.3. Large language models in healthcare

In recent years, the field has witnessed a natural progression from traditional transformer-based models to large language models (LLMs), which offer significantly enhanced capabilities in understanding, reasoning, and generalization. LLMs, often comprising billions of trainable parameters, can perform a wide range of tasks in zero-shot settings. In the healthcare domain, LLMs are increasingly employed for clinical decision support [27–29], clinical summarization [30–32] and biomedical research [33,34], among other applications. However, given the

highly sensitive nature of medical data, robust de-identification and anonymization techniques remain essential to ensure patient privacy and regulatory compliance. Consequently, LLM-based anonymization and de-identification have become active areas of research, with a growing number of approaches proposed in the literature to achieve a trade-off between privacy and utility. One of the earliest studies on LLM-based medical text de-identification is DeID-GPT [35], which employs GPT-4 in a zero-shot prompting setting to identify and mask sensitive information in unstructured clinical notes while preserving sentence structure. The categories of protected information are derived from the HIPAA-defined list, meaning indirect identifiers are not addressed. While DeID-GPT demonstrates strong performance, its reliance on an online API raises significant privacy concerns, especially because transmitting sensitive medical data to an external API violates the privacy and regulatory constraints in the healthcare sector.

To preserve the privacy in conversational settings including medical cases, AirGapAgent [36] employs two separate LLMs before sharing anything with an external party requesting information; a MinimizerLLM to decide the set of information that fulfills the request, and a ConversationalLLM that interacts with the requester. In case the requester asks for sensitive information that the MinimizerLLM has not listed, the subject of the data is asked for consent to share this information, which degrades the autonomy of the system. Similarly, LLM-Anonymizer [37] targets the redaction of HIPAA-defined identifiers in clinical notes. Unlike cloud-based solutions, it is hosted locally, thereby enhancing data privacy by avoiding external communication. The system relies on a single LLM to detect sensitive entities, but lacks a validation mechanism to correct potential hallucinations which is an inherent risk in using generative models.

A recent comparative study [38] demonstrated that LLM-based anonymizers consistently outperform traditional anonymization methods. The study evaluated several locally deployed LLMs, some fine-tuned and others used in zero-shot settings, on clinical text anonymization tasks using a set of newly proposed evaluation metrics. Fine-tuned smaller LLMs were found to outperform larger models operating in zero-shot mode, while both categories demonstrated better performance than conventional approaches in terms of sensitive information removal and clinical utility preservation. The framework proposed in [39] introduces a multi-component anonymization pipeline that leverages LLMs to balance privacy and utility. Although not specifically designed for medical text, it is noteworthy for its innovative use of multiple LLMs, highlighting a natural progression towards more sophisticated, LLM-driven anonymization techniques. Traditional anonymization methods are employed for the initial redaction, while one LLM-based evaluator assesses residual privacy risks, and the other evaluates the preservation of task-relevant utility. These components interact iteratively, refining the text through feedback loops before producing the final anonymized output.

2.4. Towards agentic LLM framework and research gaps

Despite significant progress in medical anonymization and LLM-based clinical applications, current systems typically address components such as data retrieval, anonymization, and question answering in isolation. Existing methods either rely on a single LLM or treat privacy-preserving tasks separately from reasoning and generation. Moreover, even the most recent frameworks lack fine-grained control over privacy rules or do not support reversible anonymization. To the best of our knowledge, no prior work offers an end-to-end, autonomous pipeline that integrates context generation, anonymization, public LLM prompting, and reidentification into a unified architecture. Our proposed solution fills this gap through an agentic framework that decomposes the pipeline into specialized agents, each assigned clear privacy boundaries. Local LLMs are used to handle all privacy-sensitive tasks, including structured data retrieval, narrative generation, and anonymization, within a secure environment. Only anonymized narratives are sent to public LLMs, ensuring that no raw patient data is exposed beyond the system's trust boundary. Additionally, our framework supports reversible anonymization via SQL-based identity mapping, enabling accurate deanonymization for internal use without undermining external privacy. Unlike token-level redaction or masking, our system leverages local generative models to produce semantically coherent anonymized narratives, improving both data utility and downstream LLM performance. This design represents a novel, privacy-aware approach to LLM orchestration in healthcare, one that supports modularity, auditability, and scalable deployment for sensitive clinical applications.

3. System design

This section presents the design of the proposed agentic LLM framework. It explains the system architecture, defines the responsibilities of each agent, outlines the control logic of the orchestrator, and describes the flow of data through the pipeline.

3.1. System overview

We propose a modular, agent-based question-answering system tailored for healthcare, designed to assist professionals in generating diagnostic and treatment recommendations using public LLMs. The overall architecture is illustrated in Fig. 2. The system integrates seamlessly with a hospital's EMR system via an SQL database and adheres to strict privacy standards through multi-stage anonymization and reidentification processes. A healthcare provider initiates a request through a user interface (UI) by providing a natural language question and a specific patient ID from the hospital's EMR. This request is forwarded to an orchestrator agent, which plans and breaks down the task into sub-tasks to be delegated to a set of callable agents. The orchestrator delegates the first task to an SQL agent, which performs three operations: (1) it queries the hospital's relational EMR database to retrieve structured data for the specified patient across multiple tables (e.g., Patients, Visits, Diagnoses, Medications, LabResults), and (2) it removes personal identifiers from the question to prepare it for further processing, and (3) it maps the patient ID to a local pseudonymous identifier used for future deanonymization.

The retrieved structured data is passed to a context generation agent, which transforms it into a coherent clinical narrative by capturing all medically relevant patient information. The narrative is then sent to an anonymization agent. This agent ensures compliance with privacy standards by removing any potentially identifiable information from the narrative, following a set of anonymization guidelines. This step is critical, as the narrative will be shared with a public LLM, which may log queries and responses for model improvement. The anonymized narrative will then be reviewed, evaluated, and corrected if necessary by a privacy-utility validator agent. Once the narrative is fully anonymized, it is submitted with the anonymized question through the public LLM proxy agent to a public LLM, which generates a clinical response, such as a diagnosis or treatment plan. Since the response may reference anonymized placeholders, it is passed to a deanonymization agent, which uses the local identifier mapping created in the first step to restore the patient-specific references. This final re-identified response is returned to the orchestrator, which forwards it to the UI for clinical interpretation and use by the healthcare provider.

3.2. System components

This section delineates the orchestrator's execution logic and the roles and responsibilities of each agent within the proposed agentic framework. The system is designed with modular agents coordinated by an orchestrator, enabling privacy-preserving, LLM-assisted clinical decision support. The system comprises six distinct agents and an orchestrator. Each agent is responsible for a specific, isolated task in the pipeline and can be reused or replaced with minimal impact on the overall framework.

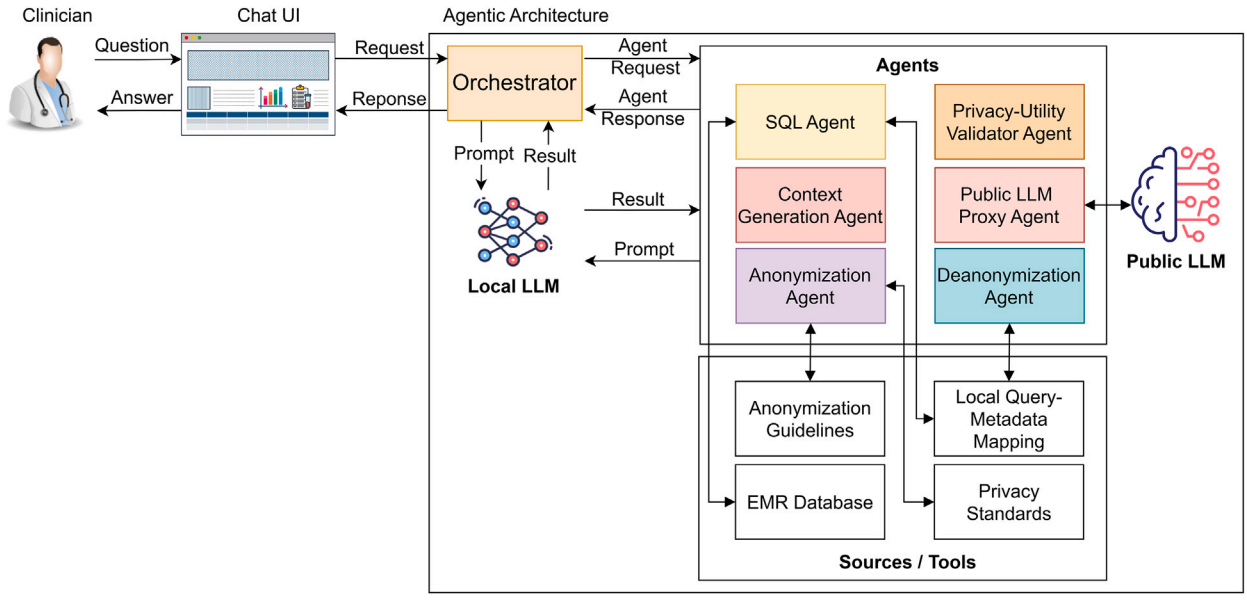


Fig. 2. Proposed contextual integrity-aware agentic LLM framework for anonymizing healthcare information.

```
1. Receive the user's question and patient ID.
2. Invoke the SQL agent to fetch data and rephrase the question.
3. Pass the SQL data to the context generation agent.
4. Forward the narrative to the anonymization agent.
5. Send the original and anonymized narratives along with the SQL data to the
    privacy-utility validator.
6. Submit anonymized narrative + question to the public LLM via the proxy agent.
7. Receive response and pass it to the deanonymization agent.
8. Return the final, reidentified response to the UI.
```

Fig. 3. Orchestrator high-level control logic.

3.2.1. *Orchestrator*

The orchestrator controls the execution of the question-answering pipeline through agent calls tailored to the clinical workflow: from processing the input question to SQL data retrieval, context generation, anonymization, privacy-utility evaluation, public LLM querying, and deanonymization. The orchestrator also maintains temporary in-memory storage to manage intermediate outputs between agents. This memory is structured to minimize the exposure of personally identifiable information (PII); each agent receives only the data necessary to complete its task, even though all components (except for prompts generated by the public LLM proxy) are executed locally. This design helps reduce potential data leakage and enforces strict internal data isolation principles. The high-level control logic executed by the orchestrator is shown in Fig. 3.

Although our implementation adopts a fixed orchestration pipeline, we also explore the feasibility of dynamic orchestration using LLMs. Specifically, instead of providing the steps that should be followed to execute the task, we test orchestration using a local LLM to generate an action plan based on task descriptions. This agent dynamically selects tools (i.e., agents) and determines the execution order through natural language reasoning. The LLM-generated plan was valid and aligned well with our fixed orchestration logic. This demonstrates that dynamic orchestration is a viable extension of our framework, either through existing library implementations or custom-designed agent interfaces. The LLM generated plan is structured as a sequence of reasoning and action steps. Each `Thought` represents the LLM's internal reasoning about what to do next. `Action` specifies the name of the tool (i.e., agent) to be invoked. `Action Input` provides the input to that tool. `Observation` represents the tool's return value, which is then used to guide the next step in the plan. This structured approach allows the agent to reason step-by-step and adapt to observations as the task progresses. Fig. 4 shows an example plan produced by a local LLM.

3.2.2. *SQL agent*

The SQL agent is the first component invoked after receiving a user-generated question from the clinical interface. These questions are often unstructured, conversational, and may include spelling or grammatical errors, informal phrasing, or ambiguous references. Importantly, they may contain multiple identifiers or domain-specific terms that can be misinterpreted, such as "Addison", "Wilson", or "Hunter" which could mistakenly be parsed as referring to a patient name Addison rather than a disease/syndrome. To accurately interpret such input, the SQL agent prompts a local LLM to extract the correct patient ID and paraphrase the question in a privacy-preserving format. This step avoids naive string redaction methods, which would either expose too much information or unintentionally strip essential clinical context. Once the patient ID is extracted, the agent uses it to perform a structured SQL query across the EMR database, joining relevant data to retrieve comprehensive patient information. In parallel, the agent creates a local, pseudonymous mapping between the true patient ID and an internal query ID. This mapping is stored temporarily and used later during the deanonymization process, ensuring that patient identity is protected throughout the intermediate stages of processing.

```
Thought: I need to fetch the patient's data first.
Action: FetchPatientData
Action Input: ----

Observation: ----

Thought: Now that I have the patient's data, I will generate a narrative from
    this data.
Action: GenerateNarrative
Action Input: ----

Observation: ----

Thought: The next step is to anonymize the generated narrative.
Action: AnonymizeNarrative
Action Input: ----

Observation: ----

Thought: The anonymized narrative needs to be validated to ensure it is useful
    clinically.
Action: ValidateAnonymization
Action Input: ----

Observation: ----

Thought: The next step is to send the validated anonymized question to the
    public LLM to get an answer.
Action: PublicLLMProxy
Action Input: ----

Observation: ----

Thought: I now know the final answer
Final Answer: ----
```

Fig. 4. Orchestrator high-level control logic.

3.2.3. Context generation agent

This agent transforms the structured SQL output into a coherent clinical narrative that integrates diagnoses, medications, lab results, and visits into a readable format. A local LLM is employed to generate this narrative, leveraging its contextual understanding to link related data points and present them in a human-readable, case-like summary. Special attention is given to preserving the chronological order of events such as clinical visits, diagnoses, treatments, and lab observations, which is essential for capturing the evolution of the patient's condition—whether showing improvement, stability, or deterioration. During the development phase, it was noticed that the local LLM sometimes miscalculates the age of the patients, which is sometimes essential to evaluate the clinical case correctly. To that end, a function to calculate the age is added to the agent to avoid such confusion. The correct age is appended to the structured SQL data to be included in the narrative. Once the narrative is generated, it is returned to the orchestrator for downstream processing.

3.2.4. Anonymization agent

This agent performs one of the most critical tasks in the system, which is transforming sensitive clinical narratives into anonymized versions that preserve diagnostic utility and clinical readability, while minimizing the risk of patient re-identification. Unlike traditional anonymization methods that rely on surface-level redaction or masking (e.g., replacing sensitive information with "[REDACTED]"), this agent leverages a multi-step, LLM-driven process to perform context-aware anonymization and paraphrasing. This approach is particularly important in clinical scenarios where readability, timeline coherence, and semantic clarity must be preserved.

The anonymization process begins with an LLM analyzing the context-rich narrative to identify all personally identifiable information (PII), guided by a predefined set of system-specific rules. These include not only patient names and contact details, but also subtle identifiers such as locations, birthdates, and even certain lab or visit dates that could indirectly lead to re-identification. The LLM's ability to interpret context allows it to address corner cases that traditional rule-based systems might miss, especially where the same token might serve as a name or a medical term depending on context. Once the identifiers are determined, the second step of the process transforms the narrative in a way that removes or generalizes sensitive elements without degrading the narrative quality or clinical coherence. For instance, dates are not simply redacted but shifted or generalized, maintaining the correct chronological relationships between clinical events. This temporal anonymization is crucial for downstream reasoning about disease progression, treatment response, or symptom evolution. Locations may be broadened, where instead of explicitly mentioning the name of the clinic, a more general geographic location is used to preserve geographic cues while obscuring exact identities, which is sufficient for clinical utility. Doctor names receive special treatment. In conventional systems, these would often be redacted using known name lists, but such methods typically leave behind contextual cues like professional titles or specialties that can easily be linked back to the individual, and consequently the patient. In contrast, this agent replaces doctor references with neutral roles that preserve function but not identity, such as "a doctor" or "a cardiology specialist".

In some cases and based on the user's preferences, real names are replaced with pseudonyms. In such case, it is important to explicitly

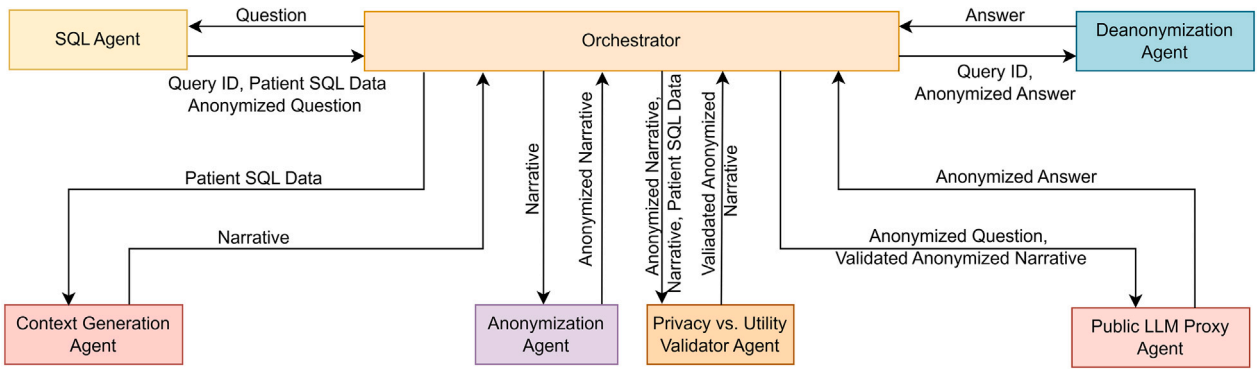


Fig. 5. Data flow diagram illustrating the interaction between the orchestrator and individual agents in the proposed agentic LLM framework.

instruct it to use internationalized names, rather than defaulting to culturally similar alternatives. Without this constraint, models tend to substitute names within the same cultural or regional context, for example, replacing a western name with another western name. This behavior risks retaining culturally identifiable markers that could aid in re-identification, particularly when combined with other contextual clues. Instead, prompting the LLM to use generalized references or culturally diverse names helps reduce this bias and further anonymize the content.

Given the risk of hallucinations and misrepresentation by generative models, a third LLM is used to validate the anonymized narrative. This validation step relies on two key inputs; the original narrative before anonymization and the anonymized version produced by the system. By comparing these two, the validation model ensures that no clinical information has been distorted or lost, that sensitive identifiers have been appropriately transformed or removed, and that the clinical utility of the narrative remains intact. The model also verifies the consistency of event sequencing and temporal coherence throughout the narrative. Overall, this agent eliminates the need for assembling multiple conventional NLP tools by using LLMs to handle identifier detection, anonymization transformation, and quality validation. The result is a streamlined, privacy-preserving pipeline that generates readable, clinically useful narratives without compromising sensitive data.

3.2.5. Privacy-utility validation agent

The privacy-utility validator Agent is responsible for performing a deeper evaluation of the anonymization process, ensuring that patient privacy is preserved without compromising the clinical value of the data. This agent leverages a local LLM to assess the quality and fidelity of the anonymized narrative using three key inputs: the original structured SQL data, the narrative generated from it, and the corresponding anonymized version. By referencing the SQL data, the agent can detect omissions or hallucinations introduced during narrative generation or anonymization. Comparing the raw and anonymized narratives allows the agent to verify that sensitive identifiers have been appropriately handled, while ensuring that the anonymization has not obscured medically relevant context. The agent also evaluates the consistency and interpretability of the anonymized output. If any privacy or utility issues are detected, the agent corrects them directly and returns an updated anonymized narrative. Otherwise, it returns the narrative unchanged.

3.2.6. Public LLM proxy agent

The public LLM proxy agent serves as the interface between the secure, local processing environment and the external public LLM. It receives two inputs; the anonymized patient narrative and the anonymized version of the clinical question. The agent constructs a prompt and submits it to a public LLM. A public LLM is selected to answer the questions for its broader training corpus, including up-to-date medical literature and clinical guidelines, and also for its superior reasoning and diagnostic capabilities compared to local models. Since the input narrative has been anonymized with high precision, the system ensures that no personally identifiable information is exposed during this external interaction. Once the public LLM returns its response, the anonymized output is sent back to the orchestrator.

3.2.7. Deanonymization agent

This agent was included to address scenarios where multiple prompts are sent to the public LLM at once. Upon receiving the anonymized responses, it is necessary to map each response back to its original record. The agent leverages the local mapping created earlier by the SQL agent to securely retrieve the patient's personal information from the EMR database using the query ID supplied by the orchestrator. It then prompts a local LLM to re-inject these identifiers appropriately into the anonymized response. Finally, the deanonymized response is sent back to the orchestrator.

3.3. Data flow

Fig. 5 shows the interactions among the orchestrator and agents in the proposed agentic LLM framework. Arrows in the figure are annotated with the input and output data to and from each agent. It is clear that data with PII is only shared with distinct locally hosted agents, protecting the privacy of the subject patient. Further details on the communication between the orchestrator, agents, and corresponding tools are depicted in Fig. 6.

4. Agent design and implementation

The proposed agentic framework is implemented with a strong emphasis on modularity, using LangChain's [40] `RunnableSequence` to orchestrate a series of agents, each defined as a `Runnable` with a well-specified set of inputs and outputs. This modular architecture allows for independent updates and design modifications to any agent, provided its interface remains consistent. Each agent is responsible for a distinct task, which simplifies both implementation and performance evaluation. All agents implement a standardized `invoke()` method, which serves as the primary interface for external communication. This method receives a structured input dictionary and returns a structured output dictionary, enabling seamless integration within the overall execution pipeline. Supporting methods within each agent are used internally to perform subtasks and are encapsulated to preserve modularity. For agents that interact with LLMs, prompt engineering plays a critical role in ensuring accurate execution and consistent results. These agents are guided by carefully crafted prompts that provide clear instructions aligned with their specific roles in the pipeline. The agent structure along with its input, prompt, and output are described for each agent below. The SQL Agent is described in Listing 1, the Context Generation Agent is described in Listing 2, the Anonymization Agent is

described in Listing 3, the Privacy Utility Evaluator Agent is described in Listing 4, the Public LLM Proxy Agent is described in 4, and the Deanonymization Agent is described in Listing 6. Listing 7 shows the orchestration logic.

The agentic framework is supported by a set of resources that guide the agents, particularly the anonymization and evaluation steps. A core component is the inclusion of privacy standards such as HIPAA, which are used to define what constitutes sensitive patient information, helping the LLM identify and protect these elements within the narrative. Complementing this is a set of anonymization guidelines, emphasizing context-aware rewriting rather than traditional redaction or masking. This ensures that narratives remain readable and clinically useful while protecting sensitive information. The agents also operate on a structured hospital EMR dataset stored in a relational SQL database. While the current implementation uses a synthetic EMR, the system is designed to be agnostic to the specific format or source of clinical data. Additionally, a local query-to-patient ID mapping is maintained to enable deanonymization after the public LLM generates a response.

The proposed architecture exhibits a highly modular agentic design. Each agent is implemented as a reusable, LangChain-compatible module that can be replaced or updated independently, without affecting other modules provided the input and output interfaces remain unchanged. For instance, the anonymization agent could be swapped for one based on differential privacy, or the context generator could be upgraded with a domain-specific LLM. This modularity enables easy customization for different clinical environments, use cases, or privacy requirements.

```
class SQLAgent:
    def __init__(self, llm, emr_db_engine):
        # Initialize agent with language model and EMR
            database engine
    def ensure_mapping_table_exists(self):
        # Ensure QueryPatientMap table exists in the
            database
    def insert_query_mapping(self, patient_id):
        # Insert patient ID and return generated query
            ID
    def get_patient_data(self, patient_id):
        # Fetch structured EMR data for the patient
            from multiple tables
        # If dataset schema changes, only this
            function needs to be updated.
    def invoke(self, input: dict) -> dict:
        """
        Input: { "question": str # Contains patient ID
            and possibly other sensitive data }
        Prompt: "You are an assistant that extracts
            patient IDs from questions and removes
            personal information from the question.
            Extract patient ID from the Question {
            question}. Remove patient ID and any
            sensitive information from the question.
            Return the ID and the paraphrased question
             in the following format: {"patient_id": "
            id", "question": "anonymized question"}.
            You must not generate extra text."
        Output: { "query_id": int,
                  "anonymized_question": str, #
                      Rephrased version
                  "sql_data": dict # Structured EMR
                      data }
        """
```

Listing 1: SQLAgent Class Overview

```
class ContextGenerationAgent:
    def __init__(self, llm):
        # Initialize agent with the language model
    def calculate_age(self, dob):
        # Calculate the patient's age
    def invoke(self, input: dict) -> dict:
        """
        Input: { "sql_data": dict # Structured EMR data
             from the SQLAgent }
        Prompt: "You are a clinical data summarization
             assistant. Here is the patient's medical
            data in a structured format:{data_summary
            } and this is their correct age {age}.
            Based on this data, write a detailed,
            clear, and accurate narrative summary of
            the patient's medical history, visits,
            diagnoses, medications, and lab results.
            Return the narrative as one paragraph in a
             python dictionary in the following format
            :{"narrative": narrative}. You must not
            generate additional text. "
        Output: { "narrative": str # Natural language
            summary of the patient's medical history }
        """
```

Listing 2: ContextGenerationAgent Class Overview

```
class AnonymizationAgent:
    def __init__(self, llm):
        # Initialize agent with the language model
    def invoke(self, input: dict) -> dict:
        """
        # Step 1: Detect PHI items
        # Step 2: Anonymize detected PHI
        # Step 3: Validate the anonymized narrative
        Input: { "narrative": str # Natural language
            summary of patient data }
        Prompt: "You are a clinical anonymization
            assistant. Narrative:{narrative}.
            Sensitive items to anonymize: {
            detected_phi}. Guidelines: {
            anonymization_guidelines}. Anonymize the
             narrative based on the provided
            anonymization_guidelines, given the list
            of detected_phi. Return the anonymized
            narrative as a python dictionary:{"
            anonymized_narrative": "..."}. No
            additional explanation.
        Output: { "anonymized_narrative": str # PHI-
            anonymized version of the input narrative
            }
        """
```

Listing 3: AnonymizationAgent Class Overview

```
class PrivacyUtilityEvaluatorAgent:
    def __init__(self, llm):
        # Initialize agent with the language model
    def invoke(self, input: dict) -> dict:
        """
        Input: {
            "anonymized_narrative": str, # Output from
                the anonymization agent
            "original_narrative": str, #
                ContextGenerationAgent output
            "sql_data": dict # Structured EMR data for
                consistency check
        }
```

```
        Prompt: "You are a privacy and utility
             evaluation assistant for clinical
             narratives. Your task is to: (1) Evaluate
             whether the anonymized narrative protects
              patient privacy and maintains clinical
             utility. (2) Use the structured data as
             ground truth to check factual consistency
             . (3) If the anonymized version is good,
             return it as-is. (4) If privacy risks or
             utility loss are found, return a corrected
              version that better preserves both. (5)
             Return your response as a python
             dictionary in the following format:{"
             final_narrative": "..."}. Here is the
             structured data (SQL): {structured_sql}.
             Original narrative: {original_narrative
             }. Anonymized narrative: {
             anonymized_narrative}. Do not generate
             any extra text."
        Output: { "final_narrative": str # Verified or
             improved anonymized version }
        """
```

Listing 4: PrivacyUtilityEvaluatorAgent Class Overview

```
class PublicLLMProxyAgent:
    def __init__(self, llm):
        # Initialize agent with a public LLM
    def invoke(self, input: dict) -> dict:
        """
        Input: {"question": str, # Doctor's query
            "anonymized_narrative": str # Anonymized
                 patient summary }
        Prompt: "You are a clinical assistant. A doctor
              has asked a question about a patient. Use
             the following anonymized narrative to
             answer the question accurately and
             clearly. Question {question}. Anonymized
             narrative {anonymized_narrative}. Return
              your response as a python dictionary in
             this format: {"answer": "..."}. You must
             not generate extra text."
        Output: {"answer": str # Response to the
             question based on anonymized narrative }
        """
```

Listing 5: PublicLLMProxyAgent Class Overview

```
class DeAnonymizationAgent:
    def __init__(self, llm, db_engine):
        # Initialize agent with language model and
             database engine
    def get_patient_id_from_query_id(self, query_id)
         :
        # Retrieve PatientID from the QueryPatientMap
             table using QueryID
    def get_patient_info(self, patient_id):
        # Fetch patient information from Patients
             table by PatientID
    def invoke(self, input: dict) -> dict:
        """
        Input: {"query_id": int, # Query identifier
               "answer": str # Anonymized response
                    from public LLM }
        Prompt: "You are a helpful assistant that
             deanonymizes clinical responses using
             patient information. Patient data: {
             patient_data}. Anonymized response: {
             anonymized_response}. Rephrase the
             response to reinsert the patient's real
             name and other identifying details when
             appropriate, based only on the
             information provided. Return your
             response as a python dictionary as follows
             : {"response": "..."}. You must not
             generate any other text."
        Output: { "deanonymized_response": str #
             Response with identifying info reinserted
              }
        """
        pass
```

Listing 6: DeAnonymizationAgent Class Overview

```
# Define all agents: SQLAgent,
     ContextGenerationAgent, AnonymizationAgent,
     PrivacyUtilityEvaluatorAgent,
     PublicLLMProxyAgent, and DeAnonymizationAgent
# Define the orchestrator chain (linear sequence)
orchestrator_chain = (
    sql_agent
    | context_generation_agent
    | anonymization_agent
    | privacy_utility_evaluator
    | public_llm_proxy_agent
    | deanonymization_agent
)
# Invoke with user-provided question from chat UI
input_question = {"question": user_input}
output = orchestrator_chain.invoke(input_question)
print(output["answer"])
```

Listing 7: Definition of Orchestration Logic

5. Use cases

There are several use cases for the proposed agentic framework, each demonstrating how it can support privacy-preserving interactions between sensitive data and powerful public LLMs. The primary focus of this paper is on clinical question answering systems, where medical professionals can query patient-specific information while ensuring compliance with privacy protection standards. However, the modular and generalizable design of the framework allows it to be adapted for other applications as well, including cohort-based research analytics, automated case summarization, and tracking and analyzing a patient's health data over time. Two detailed examples will be provided here to highlight how the framework operates across different contexts, the first use-case is the question-answering system which is the central case discussed throughout the paper, and the second one is the case related to research analytics.

5.1. Medical question-answering use case

In this use case, a doctor might want to seek recommendations regarding a patient's case from a public LLM. For example, the doctor might input the following question:

> *What is your treatment plan suggestion for Daniel Figueroa, holding ID number 4?*

Submitting this question together with a detailed description of the patient's case to a public LLM violates internationally defined privacy laws, as it exposes the patient to the risk of data leakage. Below is an example of a narrative explaining the patient's case.

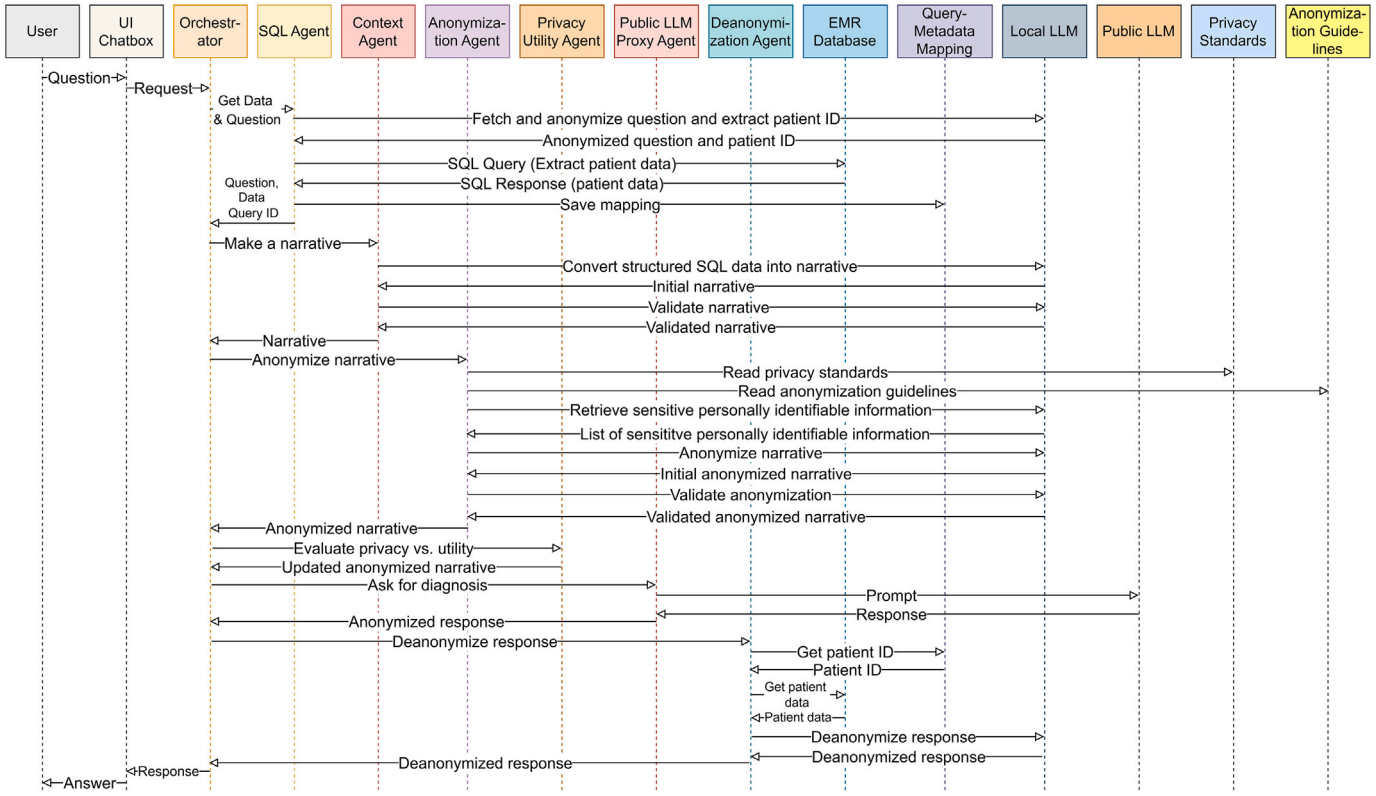


Fig. 6. Sequence diagram depicting detailed interactions between the agents and tools in the proposed agentic LLM framework.

> *Daniel Figueroa, a 33-year-old patient, had a routine check-up with Dr. Jesse Guzman on 2022-04-21. The visit resulted in a diagnosis of Hyperlipidemia. Laboratory results from the visit included a Cholesterol level of 320 mg/dL and a Glucose level of 276 mg/dL.*

In some cases, entities may require explicit patient consent, which can make patients reluctant to share their concerns freely and comfortably. Our agentic anonymization framework alleviates these concerns by anonymizing any sensitive information in both the question and the narrative before exposing it to externally hosted models. An example of the anonymized question and narrative is provided below.

> *What is your treatment plan suggestion for this patient?*
>
> *A patient in their early 30s visited the hospital for a routine checkup. Lab results revealed elevated Cholesterol and Glucose levels (320 mg/dL and 276 mg/dL respectively) which resulted in a Hyperlipidemia diagnosis.*

The anonymized narrative along with the anonymized question are submitted to a public LLM, which provides an anonymized response as shown in the example below.

> *The patient will begin a comprehensive treatment plan addressing both hyperlipidemia and elevated glucose levels. A high-intensity statin, such as atorvastatin 40 mg daily, will be initiated to lower cholesterol, alongside metformin 500 mg twice daily to manage hyperglycemia, pending confirmation of diabetes with an HbA1c test. Lifestyle modifications including a low-saturated-fat diet, increased physical activity, and weight management will be strongly encouraged. Follow-up labs will be scheduled in 8 weeks to monitor lipid and glucose levels and adjust therapy as needed.*

The deanonymization agent then uses a secure local mapping to reinject identifiers wherever needed, ensuring the final answer is both informative and personalized. By controlling what data leaves the secure environment and reinserting identifiers only at the endpoint, this use case demonstrates how the system enables practical, LLM-powered decision support without compromising patient privacy.

5.2. Cohort-based research analytics use case

Another compelling use case for the proposed agentic framework is in supporting retrospective clinical research, particularly cohort-based studies. For instance, a researcher may wish to analyze patterns across a group of patients who share a specific condition. A possible question could be:

> *What are the common treatment pathways and outcomes for patients diagnosed with Type 2 Diabetes who were also prescribed metformin within the first three months of diagnosis?*

In this scenario, the orchestrator would execute its workflow for a batch of cases, using the patients' diagnosis as a search key to retrieve relevant EMRs from the database. Each EMR then goes through the anonymization process and the batch is finally submitted to the public LLM which will recognize any patterns answering the researcher's question. If those observations are to be re-injected into the EMR database, deanonymization could be executed using the established local mapping. The result is a scalable, privacy-aware system that enables rich qualitative insights from patient data without compromising confidentiality, which is a critical requirement for modern clinical research involving sensitive health records.

6. Evaluation and results

This section presents a comprehensive evaluation of the proposed framework, focusing on its experimental setup, local model selection, and anonymization performance. Both quantitative and qualitative

Table 1
Relational Schema of the Synthetic EMR Dataset.

Table	Primary Key	Foreign Keys	Key Attributes
Patients	PatientID	–	FirstName, LastName, DateOfBirth, Gender, ContactInfo
Visits	VisitID	PatientID	VisitDate, ReasonForVisit, DoctorName
Diagnoses	DiagnosisID	VisitID	ICD10Code, Description
Medications	MedicationID	VisitID	DrugName, Dosage, Duration
LabResults	LabResultID	VisitID	TestName, ResultValue, Units, ReferenceRange

analyses are conducted to benchmark the system against established NER tools and to assess its effectiveness in handling context-sensitive clinical narratives.

6.1. Experimental setup

The system is designed for local deployment within hospital or medical entity environments to ensure maximum data privacy. Real patient data never leaves the local infrastructure, preventing exposure to unauthorized external parties. The only web-based interaction occurs when querying a public LLM for clinical assistance, which is strictly performed on already anonymized data that has undergone multiple privacy validation iterations locally, via carefully prompted local LLMs. The experiment was run on a cloud VM, accessed through Google Colab Pro, which hosts a 27 GB system RAM and a single NVIDIA L4 Tensor Core GPU with 24 GB VRAM. The agentic LLM framework was developed in Python leveraging the LangChain framework and an SQL database engine. A user-friendly chat interface is implemented using Gradio.

6.1.1. Synthetic dataset construction

A synthetic electronic medical records (EMR) dataset was generated to simulate patient demographic information, clinical visits, diagnoses, medications, and laboratory results. Synthetic data was selected due to the sensitive nature of downstream healthcare applications and to avoid exposing real patient records. Moreover, synthetic data enables controlled and reproducible experimentation while preserving complete knowledge of the ground truth for evaluation purposes.

The dataset is stored in an SQLite relational database, allowing the proposed framework to operate on structured clinical data in a manner consistent with typical hospital EMR systems. Table 1 presents the relational schema of the dataset. The database consists of five interconnected tables: *Patients*, *Visits*, *Diagnoses*, *Medications*, and *LabResults* and is designed to mirror a realistic EMR structure with longitudinal patient histories. The dataset contains 100 patients, 242 clinical visits, 350 diagnoses, 348 medication records, and 432 laboratory results. This facilitates multi-condition and multi-visit narrative generation. The dataset includes both direct identifiers and quasi-identifiers consistent with HIPAA-defined PHI categories, including patient first and last names, dates of birth, visit dates, doctor names, contact information, gender, and structured identifiers such as patient IDs. Because all narratives are generated directly from the structured SQL data, these identifiers constitute the complete ground truth for anonymization evaluation.

Importantly, the synthetic records were not generated purely at random. Instead, structured generation instructions were provided to a large language model (ChatGPT), which populated the database tables according to predefined constraints and relational consistency rules. This controlled generation process ensured coherence across tables — for example, maintaining consistency between patient demographics, visit histories, diagnoses, medications, and laboratory results — thereby producing clinically plausible longitudinal records while preserving full traceability of inserted PHI elements. Although synthetic, the dataset preserves realistic EMR relationships and temporal progression, enabling rigorous evaluation of context-aware anonymization under multi-visit, multi-condition clinical scenarios.

This approach allows reproducible experimentation without exposing real patient information to privacy leakage. The framework is dataset-agnostic and can be adapted to other EMR databases as needed. Additionally, a local query–patientID mapping table supports the deanonymization functionality. The system employs both local and public LLMs. The local LLM is Qwen (Qwen2.5-7B-Instruct), loaded via `huggingface transformers` library and integrated through LangChain wrappers to perform core tasks such as narrative generation, anonymization, and privacy evaluation. The public LLM used is OpenAI's ChatGPT (gpt-4o), accessed through API calls, which handles clinical question answering based on anonymized narratives. Different LLM models and versions can be used at the discretion of the user. Prompts were carefully designed to emphasize adherence to anonymization guidelines and hence enhance the privacy of the proposed agentic architecture.

6.2. Local LLM selection

To identify the most suitable local LLM for our application, without requiring fine-tuning or retraining, we conducted an evaluation using 100 synthetically generated clinical questions. Each LLM was given a fixed prompt instructing it to anonymize the input by removing sensitive information and correctly extracting the patient ID for downstream processing. A successful outcome was defined as accurately identifying and removing the patient ID from the question, preferably with appropriate paraphrasing to maintain coherence. Five local LLMs were tested: google/flan-t5-base, Qwen/Qwen2.5-7B-Instruct, meta-llama/Meta-Llama-3-8B-Instruct, meta-llama/Llama-2-7b-chat-hf, and mistralai/Mistral-7B-Instruct-v0.2. The performance varied significantly: flan-t5-base failed in all cases, unable to anonymize even simple one-sentence questions. The Mistral and LLaMA models produced limited successful outputs but frequently hallucinated or mishandled the task. Only Qwen/Qwen2.5-7B-Instruct achieved a 100% success rate, consistently anonymizing the input and extracting the correct patient ID. Based on these results, Qwen was selected for use in our framework.

6.3. Evaluation metrics

To evaluate the performance of the proposed approach, we use standard classification metrics including Precision, Recall, and F1 Score. Due to the lack of reliable, generalized anonymization benchmarks, the generated narratives were manually annotated. Items deemed as direct or indirect PII were labeled as sensitive, while medically relevant content such as diagnoses, medications, and symptoms were labeled as non-sensitive. Based on this manual labeling, we identified true positives (correctly anonymized sensitive information), false positives (non-sensitive content mistakenly anonymized), and false negatives (missed sensitive information). These counts were used to compute evaluation metrics for our method and compare its performance against five widely used NER-based anonymization techniques as summarized in Table 2.

6.4. Anonymization quantitative results

This section presents a quantitative comparison between our proposed agentic LLM framework and five widely used NER techniques: spaCy, Stanza, Flair, a transformer-based BERT model, and Presidio.

Table 2
Tools used for comparison to our proposed agentic LLM framework.

Tool	Domain	Technique
spaCy [41]	General-purpose	Statistical + Rule-based (CNNs, feature-based)
Stanza [42]	Biomedical/General	BiLSTM + CRF (trained on biomedical datasets like i2b2 [43])
FLAIR [44]	General-purpose	Character-level LSTM + CRF; supports stacked embeddings
bert-base-NER [45]	General-purpose	Transformer (BERT-base fine-tuned on CoNLL-2003 [46])
Presidio [47]	General-purpose	Rule-based + Machine learning (Microsoft PII detection tool)

Table 3
Evaluation Metrics for Anonymization and NER Techniques.

Method	TP	FP	FN	Precision	Recall	F1 Score
spaCy	686	1397	82	0.329	0.893	0.481
Stanza	686	1318	82	0.342	0.893	0.495
Flair	305	287	463	0.515	0.397	0.449
Transformers	380	707	388	0.350	0.495	0.410
Presidio	688	1010	80	0.405	0.896	0.558
Ours	**719**	**0**	**49**	**1.000**	**0.936**	**0.967**

Table 4
Statistical stability analysis on synthetic dataset across multiple runs.

Category	Metric	Mean	Std
Privacy	Recall	0.971	0.008
Utility	Medication Preservation	0.974	0.008
	Dosage Preservation	0.769	0.026

As shown in Table 3, spaCy, Stanza, and Presidio achieve high recall but suffer from low precision. This indicates their tendency to identify most sensitive PII correctly (high true positives, low false negatives), but at the cost of mistakenly classifying many non-sensitive elements as PII (high false positives). In the medical domain, such over-removal of information significantly reduces data utility, often rendering clinical narratives incomplete and less effective for downstream tasks such as LLM-based treatment recommendation. In contrast, Flair and BERT-base exhibit relatively balanced but generally low precision and recall, suggesting a high number of both false positives and false negatives. This level of performance is insufficient for reliable anonymization, as it compromises both privacy and utility. Our proposed approach outperforms all baseline methods, demonstrating superior precision and recall, and thereby achieving a more effective balance between privacy preservation and narrative utility in clinical texts.

To assess the statistical stability of the proposed framework, we conducted multiple runs on the synthetic dataset using ground-truth annotations of sensitive identifiers. This enables a deterministic and objective assessment of anonymization performance without relying on subjective annotation. Given that the anonymization process may involve paraphrasing rather than explicit removal, we focus on recall as the primary metric, defined as the proportion of sensitive identifiers successfully removed from the generated narratives. The results, summarized in Table 4, demonstrate consistently high performance with minimal variability across runs.

The framework achieves a recall of 0.971 with a low standard deviation (0.008), indicating that sensitive identifiers are reliably removed despite the stochastic nature of LLM-based generation. In terms of utility preservation, medication mentions remain largely intact (0.974 ± 0.008), suggesting that core clinical content is preserved through the anonymization process. Dosage information shows a lower preservation rate (0.769 ± 0.026), reflecting the fact that dosage values are more likely to be generalized or partially removed during anonymization due to their higher specificity and potential to contribute to re-identification risk. The balance between preserving detailed dosage information and protecting privacy can be further adjusted depending on application requirements, for example by modifying anonymization constraints or prompting strategies. Overall, the low variability across all metrics confirms that the proposed pipeline produces stable and reproducible outputs, while maintaining a consistent balance between privacy protection and clinical utility.

To further contextualize our results within recent LLM-based anonymization approaches, we compare our framework with representative methods in the literature. The LLM-Anonymizer framework [37] employs a local LLM to detect and remove sensitive information, achieving a removal success rate of approximately 98%–99% at the character level. However, its scope primarily focuses on direct identifiers defined under HIPAA, such as names, dates, and locations, without explicitly addressing quasi-identifiers that may contribute to re-identification risk.

RUPTA [39] introduces a utility-preserving anonymization approach that incorporates iterative LLM-based evaluation of both privacy and utility. While this method shares a similar objective of balancing anonymization with usability, its formulation remains centered on prompt-based optimization and evaluation of sensitive content, without explicitly modeling quasi-identifiers as discrete attributes.

In contrast, our framework adopts a modular, agentic design that decomposes the anonymization process into multiple coordinated stages, including context generation, anonymization, validation, and utility refinement. Importantly, our approach explicitly models quasi-identifiers and quantifies their presence across anonymization stages, enabling a more fine-grained assessment of re-identification risk. This allows for iterative correction of residual privacy leaks while supporting controlled reintroduction of clinically relevant information.

The LLM-based techniques we included rely on locally deployed language models. Proprietary public LLMs such as GPT-4 are excluded from experimental comparison due to possibility of private information leakage during prompt transmission which violates privacy constraints in clinical environments.

6.5. Clinical consistency and hallucination analysis

To evaluate the reliability of the LLM-based agents, we analyze clinical consistency by comparing generated narratives (before anonymization) against the underlying structured data in the synthetic dataset. Consistency is defined as the proportion of clinical attributes correctly preserved in the generated narrative, including diagnoses, medications, and dosage values. The results show high consistency for diagnoses (98.4%) and medications (97.3%), indicating that the framework preserves core clinical information with minimal distortion. Importantly, we do not observe any instances where new diagnoses or medications are introduced in the generated narratives. All discrepancies correspond to missing values rather than incorrect or fabricated clinical information.

Dosage preservation is lower (68.9%), reflecting that dosage values are occasionally omitted during generation. Unlike diagnoses and medications, which are consistently retained, dosage information is more sensitive to omission in the current prompting setup. This behavior can be mitigated by explicitly emphasizing the retention of dosage values in the generation prompt. Overall, these results demonstrate that the proposed framework maintains high factual consistency, with no evidence of hallucinated clinical entities, while preserving the majority of clinically relevant information in the generated narratives.

Original narrative with sensitive items highlighted

Hunter Lewis, a 33-year-old school teacher from Birmingham, UK, was brought to Queen Elizabeth Hospital after experiencing confusion and slurred speech during a parent-teacher meeting. His wife reported recent complaints of headache, personality changes, and insomnia. Neurological examination and MRI ruled out acute stroke. Labs showed mild hyponatremia and elevated liver enzymes. Dr. Emily Rashid (Neuropsychiatry) diagnosed him with autoimmune encephalitis, suspected post-viral. He was started on Methylprednisolone IV followed by Mycophenolate mofetil as an immunosuppressive. A lumbar puncture confirmed oligoclonal bands. Family history was non-contributory, though Hunter had a distant cousin with lupus. Follow-up neuropsychological testing revealed executive dysfunction. He is currently undergoing outpatient cognitive therapy and immune modulation. The case emphasizes the subtle onset of neuroimmune disorders in young adults. Despite improvement, Hunter continues to struggle with focus and speech under stress. Early intervention likely prevented permanent deficits.

Fig. 7. Original narrative of Case 1 with sensitive items highlighted.

6.6. Anonymization qualitative evaluation — case studies

To highlight the superior contextual understanding and anonymization capabilities of our agentic, context-aware architecture, this section presents narrative examples containing corner cases that may challenge conventional Named Entity Recognition (NER) approaches. For each case, a human annotator determines all the sensitive information in the narrative. Then, we show the corresponding anonymized narrative produced by our framework and compare it to standard NER techniques. In particular, two case studies involving the term "Hunter" are examined: one in which "Hunter" is the patient's first name, and another where the patient is diagnosed with "Hunter" syndrome. These cases are designed to test the system's ability to reason from context.

6.6.1. Case 1: "Hunter" is a patient's first name

Fig. 7 shows the original narrative describing the case of a patient named *Hunter Lewis*. Using our proposed architecture, combined with carefully engineered prompts for anonymization, the generated narrative was successfully anonymized as shown in Fig. 8. All personally identifiable information (such as patient name, true age, hospital name, doctor name, and location) as well as indirect identifiers (such as parent–teacher meeting, diagnosis of a cousin) were removed. Importantly, the anonymized narrative retained sufficient clinical detail to allow the downstream public LLM to generate an accurate and contextually relevant response. This demonstrates the effectiveness of our approach in balancing privacy preservation with clinical utility.

Table 5 summarizes the sensitive items identified by five different NER techniques for this narrative. SpaCy, Stanza, and Flair incorrectly labeled two medically relevant terms as sensitive, which compromised the clinical utility of the narrative. Additionally, these models failed to detect indirect identifiers which are details that, while not explicit, could lead to re-identification when combined (e.g., an incident involving a school teacher during a parent–teacher meeting). Notably, Flair also failed to classify the patient's age as sensitive. The transformer-based model (bert-base-NER), despite being a general-purpose model, incorrectly identified four medical terms as sensitive. The removal of these terms degraded the narrative's clinical completeness and could hinder accurate diagnosis. Lastly, Presidio correctly classified all the sensitive items it identified. However, it missed key details such as the hospital name and several indirect identifiers. While this preserved the utility of the case, it left critical privacy concerns unaddressed due to the exposure of quasi-identifiers.

6.6.2. Case 2: A patient diagnosed with "Hunter" syndrome

Fig. 9 shows a narrative where the patient is diagnosed with *Hunter Syndrome*. The term "Hunter" presents a potential ambiguity, as it can also refer to a person's name as seen in Case 1. Most traditional NER models are likely to misclassify it as a sensitive term. However, our approach leverages contextual understanding to accurately differentiate between names and medical terminology. As a result, the narrative was successfully anonymized as shown in Fig. 10 where all personally identifiable information (names, age, locations) and indirect identifiers (the mention of two other male siblings) were correctly detected and anonymized, while medically relevant terms, including Hunter syndrome, were preserved, maintaining the full utility of the case. In contrast, all other NER methods under evaluation misclassified the term Hunter as sensitive, leading to a loss of critical clinical information and reducing the narrative's utility. As in the previous case, SpaCy, Stanza, and Flair mistakenly labeled key medical terms as sensitive. While SpaCy successfully anonymized the indirect identifier, it failed to detect the patient's gender. Detailed lists of the terms identified as sensitive by each NER technique are provided in Table 6.

The qualitative results showcased through the two test cases clearly demonstrate the limitations of traditional NER systems, which frequently misclassify clinical terms as sensitive and overlook indirect identifiers, ultimately compromising both privacy and utility. In contrast, our approach, aided by precise LLM prompting and contextual understanding, successfully anonymized sensitive elements while preserving the full medical value of the narratives. The ability to distinguish between ambiguous terms like Hunter as a syndrome versus a name showcases the strength of context-aware anonymization. Furthermore, the consistent anonymization performance across both direct and indirect identifiers highlights the robustness and adaptability of our architecture. These findings underscore the significance of the reasoning, planning, and refinement iterations adopted by our agentic LLM framework, especially when utility and privacy are of high importance.

6.7. Evaluation on a public clinical dataset (MIMIC-III)

To further demonstrate the robustness of our proposed framework, we conducted additional experiments on clinical notes from the publicly available MIMIC-III dataset [48–50]. MIMIC-III consists of de-identified electronic health records of intensive care unit patients, where each record contains a free-text clinical note written by a healthcare professional and stored in the NOTEEVENTS table. For evaluation, we selected the first 100 notes from this table. Unlike our synthetic

Anonymized narrative

A 35-45 year-old individual was referred to a hospital for evaluation following a period of confusion and slurred speech during a work-related meeting. Recent reports indicated headaches, mood swings, and difficulty sleeping. Initial neurological assessment and imaging studies did not suggest an acute stroke. Laboratory tests showed mild sodium levels below normal and elevated liver function indicators. A doctor in neuropsychiatry suspected autoimmune encephalitis, possibly triggered by a virus, and initiated treatment with intravenous methylprednisolone followed by mycophenolate mofetil. Cerebrospinal fluid analysis later confirmed the presence of oligoclonal bands. Family history was unremarkable except for a distant relative with lupus. Subsequent neuropsychological evaluations highlighted issues with planning and organization. The patient is now receiving outpatient cognitive behavioral therapy and ongoing immune management. This case underscores the gradual presentation of neuroimmune conditions in middle-aged adults. Although there has been some recovery, the person still experiences concentration and verbal difficulties when under pressure. Early intervention seems to have mitigated potential long-term damage.

Fig. 8. Case 1 narrative anonymized by our proposed approach.

Original narrative with sensitive items highlighted

Ethan Miller, an 8-year-old boy from Toronto, Canada, was referred to Children's Specialty Medical Center for evaluation of developmental delay, hearing loss, and coarse facial features. His pediatrician had noted progressive abdominal distension and frequent ear infections. Genetic testing revealed a deficiency of iduronate-2-sulfatase, confirming the diagnosis of Hunter syndrome (Mucopolysaccharidosis type II). He is under the care of Dr. Natalie Hughes, a metabolic disease specialist, who initiated weekly Idursulfase (Elaprase) enzyme replacement therapy. Cardiac echocardiography showed early valvular thickening, and ENT evaluation confirmed conductive hearing loss. Neurodevelopmental testing showed borderline cognitive function, and occupational therapy has been initiated. The family is undergoing genetic counseling, as Hunter syndrome is X-linked and the couple has two other male children. Despite the progressive nature of the condition, early treatment has improved Ethan's mobility and reduced respiratory infections. The care team is monitoring for potential complications including hydrocephalus and cardiac deterioration.

Fig. 9. Original narrative of Case 2 with sensitive items highlighted.

Anonymized narrative by our proposed approach

A child was referred to a children's hospital for evaluation of developmental delay, hearing loss, and coarse facial features. The pediatrician had noted progressive abdominal distension and frequent ear infections. Genetic testing revealed a deficiency of iduronate-2-sulfatase, confirming the diagnosis of Hunter syndrome. The child is under the care of a metabolic disease specialist, who initiated weekly enzyme replacement therapy. Cardiac echocardiography showed early valvular thickening, and ENT evaluation confirmed conductive hearing loss. Neurodevelopmental testing showed borderline cognitive function, and occupational therapy has been initiated. The family is undergoing genetic counseling, as Hunter syndrome is X-linked. Despite the progressive nature of the condition, early treatment has improved the child's mobility and reduced respiratory infections. The care team is monitoring for potential complications including hydrocephalus and cardiac deterioration.

Fig. 10. Case 2 anonymized by our proposed approach.

Table 5
Case 1 — Sensitive items detected by 5 different NER approaches. TP: True Positives, FP: False Positives, FN: False Negatives.

	Spacy	Stanza	Flair	bert-base-NER	Presidio
TP	Hunter Lewis 33-year-old Birmingham UK Queen Elizabeth Hospital Dr. Emily Rashid Hunter Hunter	Hunter Lewis 33-year-old Birmingham UK Queen Elizabeth Hospital Dr. Emily Rashid Hunter Hunter	Hunter Lewis Birmingham UK Queen Elizabeth Hospital Emily Rashid Hunter Hunter	Hunter Lewis Birmingham UK Queen Elizabeth Hospital Emily Rashid Hunter Hunter	Hunter Lewis 33-year-old Birmingham UK Emily Rashid Hunter Hunter
FP	Methylprednisolone IV Mycophenolate	Neuropsychiatry Methylprednisolone IV	Methylprednisolone IV MRI	Neuropsychiatry Methylprednisolone IV Myco MR	– none –
FN	school teacher parent–teacher meeting cousin	school teacher parent–teacher meeting cousin	33-year-old school teacher cousin parent–teacher meeting	school teacher cousin 33-year-old parent–teacher meeting	school teacher Queen Elizabeth Hospital parent–teacher meeting cousin

Table 6
Case 2 — Sensitive items detected by 5 different NER approaches. TP: True Positives, FP: False Positives, FN: False Negatives.

	Spacy	Stanza	Flair	Presidio	**bert-base-NER**
TP	Ethan Miller, 8-year-old, Toronto, Canada, Children's Specialty Medical Center, Dr. Natalie Hughes, Ethan, two other male children	Ethan Miller, 8-year-old, Toronto, Canada, Children's Specialty Medical Center, Natalie Hughes, two, Ethan	Ethan Miller, Toronto, Canada, Children's Specialty Medical Center, Natalie Hughes, Ethan	Ethan Miller, 8-year-old, Toronto, Natalie Hughes, Ethan	Ethan Miller, Toronto, Canada, Children's Specialty Medical Center, Natalie Hughes, Ethan
FP	weekly, Idursulfase, Cardiac, ENT, Neurodevelop- mental, Hunter	weekly, Idursulfase, Elaprase, Hunter	Hunter, Mucopolysaccha- ridosis II, Idursulfase, Elaprase, Hunter	Hunter, weekly, Hunter	Hunter, Hunter
FN	boy	boy, male children	8-year-old, boy, two other male children	Canada, Children's Specialty Medical Center, two other male children, boy	8-year-old, boy, two other male children

EMR data, MIMIC-III narratives are unstructured and contain pre-redacted placeholder tokens (e.g., [**Doctor Last Name**], [**Hospital**], [**Date**]), which introduce additional challenges for LLM-based processing. These placeholders were treated as proxy ground truth during evaluation, and our anonymization agent was expected to remove or normalize them to improve both privacy and readability. Due to the absence of structured SQL data, the SQL agent was not required in this setting. However, the context generation (narrative) agent remained essential to normalize the input format, as redaction markers and irregular formatting negatively impacted the LLM's ability to consistently follow anonymization and formatting instructions. Furthermore, since the dataset is already de-identified, the deanonymization agent was not utilized. To handle the increased length of clinical narratives, we adopted a chunking strategy in which long documents were divided into smaller, semantically coherent segments and processed sequentially. The outputs were then aggregated to produce the final anonymized narrative. This approach ensured compatibility with LLM context limits while preserving narrative coherence. Overall, these adaptations highlight the flexibility of the proposed agentic architecture, demonstrating its ability to operate effectively across different data modalities while addressing both explicit and implicit privacy risks in real-world clinical text. To quantitatively evaluate the performance of the anonymization agent on MIMIC-III clinical narratives, we leveraged the dataset's pre-existing placeholder tokens (e.g., [**Doctor Name**], [**Hospital**], [**Date**]) as ground truth. These tokens explicitly mark protected health information that has been removed during dataset curation. For each clinical note, all placeholder tokens were automatically extracted using regular expression matching, and the corresponding anonymized outputs were analyzed to assess how these tokens were handled. Each placeholder instance was categorized into one of three outcomes: (1) *Not removed*, where the original placeholder token (including brackets) remained unchanged in the output; (2) *Partially retained*, where the bracketed token was removed but the full textual content inside the placeholder (e.g., "Doctor Name" or "2134-3-18") was preserved in the output; and (3) *Clean*, where the placeholder was successfully removed or transformed such that neither the original token nor its full textual content appeared in the output. To ensure robust matching, both placeholder content and model outputs were normalized through lowercasing and removal of punctuation prior to comparison, enabling consistent detection despite minor formatting variations. This evaluation strategy enables automatic, large-scale assessment without manual annotation, while

capturing a key failure mode of LLM-based anonymization systems, namely, the removal of placeholder markers without eliminating the underlying sensitive content.

On a subset of 100 clinical notes comprising 4557 placeholder instances, the proposed framework achieved a clean transformation rate of 98.6%, with 0.0% of placeholders left unmodified and 1.6% partially retained, demonstrating strong generalization to real-world clinical narratives. The low rate of unmodified placeholders indicates that the model effectively identifies and processes PHI markers. The partially retained cases primarily arise when placeholder brackets are removed but structured content, such as dates, synthetic identifiers, or templated name patterns, remains in the text.

It is important to note that MIMIC-III adopts a conservative de-identification strategy, where certain numerical values (e.g., medication dosages or clinical measurements) may also be replaced with placeholders. In contrast, our anonymization objective explicitly preserves clinically relevant information. As a result, some instances categorized as *partially retained* correspond to medically meaningful content that was intentionally preserved and do not necessarily represent true privacy risks.

6.8. Formal privacy and robustness analysis

To evaluate the robustness of the proposed anonymization framework, we simulate an adversarial re-identification attack. In this setting, a public LLM (acting as the attacker) is provided with anonymized clinical narratives and prompted to infer any potential patient-identifying information as follows.

> *You are a privacy attacker. Based on contextual clues in the attached anonymized narrative, try to infer any patient identifying information such as names, locations, dates, or institutions. If no information can be inferred, respond NONE.*

This setup reflects a realistic threat scenario in which an external adversary attempts to exploit residual contextual clues to reconstruct identity. We conduct this evaluation on 100 anonymized narratives derived from the MIMIC-III dataset (Section 6.7). Across all evaluated samples, direct identifiers (e.g., names, contact details, unique IDs, and exact locations) were consistently removed by the proposed framework. As a result, the adversarial model failed to correctly reconstruct any true patient identifiers. The model did not generate explicit guesses of patient identity. Instead, it only listed possible identifiers inferred from contextual clues (e.g., general categories of locations or institutions). None of these corresponded to the true underlying identifiers present in the dataset.

A small fraction of cases (1.6%) exhibited partial retention of deanonymized content, as discussed in the previous section. A closer inspection of these cases reveals that the retained elements correspond to medication dosage, few dates, or deidentified names. These elements do not provide sufficient linkage signals for re-identification and therefore do not constitute a significant privacy risk in isolation.

However, the analysis reveals the presence of quasi-identifiers, particularly in clinically detailed attributes such as medication types, dosages, and treatment plans. While these features may, in some cases, narrow the set of possible individuals, especially for rare conditions, they do not enable direct identification without additional external information. The retention of such attributes is intentional, as they are essential for preserving clinical utility and ensuring the usability of the data in downstream healthcare applications. This observation highlights an inherent privacy–utility trade-off. Stricter anonymization techniques (e.g., k-anonymity) would require suppressing or generalizing clinically relevant information, potentially reducing data value and limiting applicability in real-world clinical settings.

In addition to clinically derived quasi-identifiers, our analysis also considers several non-HIPAA quasi-identifiers that may contribute to re-identification risk in free-text clinical narratives. These include patient profession, family member professions, family structure, and patient age. While many other quasi-identifiers may be embedded in unstructured text, the open-ended nature of clinical narratives makes exhaustive enumeration impractical. Therefore, we focus on this subset and provide a detailed quantitative analysis in the subsequent section. Overall, the results demonstrate that the proposed framework effectively removes direct identifiers and prevents successful re-identification under an adversarial setting. Any residual privacy risk arises from quasi-identifiers that are inherently tied to clinically meaningful information, rather than from failures of the anonymization process itself. This limitation is well-recognized in healthcare data sharing, where preserving clinical utility necessarily introduces a controlled degree of identifiability.

6.9. Ablation study and privacy–utility trade-off analysis

While each component serves a distinct role within our proposed privacy-preserving framework, we conduct ablation experiments to quantify their individual contributions to anonymization robustness and clinical utility. The ablation study focuses on three core components of the pipeline:

1. Context Generation Agent: transforms semi-structured clinical notes into coherent narrative text.
2. Anonymization Agent: performs primary sensitive information (mainly PHI) removal.
3. Privacy-Utility Validator Agent: detects and corrects residual privacy leaks (through quasi-identifiers) and ensures that data utility is not affected.

Other components in the framework, including the SQL agent, the public LLM proxy for downstream clinical question answering, and the deanonymization agent, are not included in the ablation analysis. These components are dataset- or application-specific and primarily support system integration and usability rather than directly influencing anonymization performance.

We evaluate the system on a subset of 100 discharge summaries extracted from the MIMIC dataset. These notes are characterized by heterogeneous formatting, including bullet points, key–value structures, and partially tokenized sensitive information. Due to the unstructured nature of the dataset, we isolate the anonymization pipeline and apply it in a staged manner. Long notes are first segmented into manageable chunks to ensure compatibility with LLM context limits. We then progressively enable components of the pipeline to assess their impact as follows:

- Configuration A (Baseline): Direct anonymization of raw clinical notes
- Configuration B: Narrative generation followed by anonymization
- Configuration C (Full Pipeline): Narrative generation + anonymization + validator

We provide a combination of quantitative and qualitative measures to assess the contribution of each component as per the configurations. First, we quantify the number of successfully processed text chunks to justify the need for the context generation agent.

As shown in Table 7, the baseline configuration (Configuration A), which applies anonymization directly to raw clinical notes from MIMIC-III, achieves a chunk processing rate of 85%. In the remaining 15% of cases, the model fails to produce outputs in the required format, often due to the presence of irregular structures such as nested tokens and fragmented formatting in the original notes. In contrast, enabling the context generation agent (Configurations B and C) results in a 100% chunk processing rate. By transforming semi-structured clinical notes into coherent narrative text, the context generation agent normalizes the input format, allowing the anonymization agent to operate consistently across all segments. These results highlight the critical role of

Table 7
Ablation study results on MIMIC-III clinical notes. We report both the count and percentage of placeholder outcomes across configurations.

Configuration	Chunk Proc. (%)	Clean	Partial	Not Removed
Configuration A	85.0	4343 (95.3%)	188 (4.1%)	26 (0.6%)
Configuration B	100.0	4481 (98.3%)	74 (1.6%)	2 (0.0%)
Configuration C	100.0	4494 (98.6%)	63 (1.4%)	0 (0.0%)

Table 8
Percentage of narratives revealing quasi-identifiers (n=75)

Attribute	Original	Anonymization		Utility
		L1	L2	
Spouse profession (A1)	5.3%	4.0%	1.3%	1.3%
Age (A2)	92.0%	48.0%	25.3%	44.0%
Family structure (A3)	44.0%	14.7%	10.7%	10.7%
Patient profession (A4)	13.3%	16.0%	6.7%	6.7%

input normalization in ensuring robustness when processing real-world clinical text.

In addition to processing robustness, we evaluate anonymization effectiveness using the placeholder-based metrics introduced in Section 6.7. Specifically, we report the clean transformation rate, representing the proportion of placeholder tokens successfully removed or normalized, the partial retention rate, capturing cases where placeholder brackets are removed but the underlying content remains, and the not removed rate, where placeholder tokens persist unchanged in the output.

Table 7 summarizes the contribution of each component to anonymization performance. Configuration A, which applies anonymization directly to raw clinical notes, achieves a clean transformation rate of 95.3%, but suffers from both incomplete processing (85% chunk success rate) and higher residual errors, with 188 partially retained and 26 unprocessed placeholder instances. These errors occur due to the irregular structure of the input, including fragmented formatting and nested tokens.

Introducing the context generation agent (Configuration B) results in a substantial improvement across all metrics. The chunk processing rate increases to 100%, and the number of partially retained instances is reduced by more than half (from 188 to 74). This demonstrates that normalizing semi-structured input into coherent narrative text significantly improves the consistency and reliability of the anonymization process.

The full pipeline (Configuration C), which includes the validator agent, further improves performance by eliminating all unprocessed placeholders and reducing partial retention to 63 instances (1.4%). While the improvement over Configuration B appears incremental in aggregate metrics, this gain is primarily concentrated in resolving residual direct identifier leakage. This indicates that the validator plays a targeted role in catching edge cases that persist after initial anonymization, rather than broadly altering transformation behavior.

While these results demonstrate strong performance in removing direct identifiers, they do not fully capture more subtle privacy risks. In particular, quasi-identifiers exhibit more nuanced behavior that is not reflected in placeholder-based evaluation. This motivates a more fine-grained analysis of quasi-identifier exposure across anonymization stages.

To provide a more detailed assessment of privacy, we analyze the presence of selected quasi-identifiers in the generated narratives. Specifically, we consider four attributes: patient age, patient profession, family structure, and spouse/family member profession. These attributes are commonly recognized as quasi-identifiers in the privacy literature and may contribute to re-identification risk when combined. Table 8 shows the percentage of quasi-identifiers exposed in the narratives at different stages of the pipeline. The four columns correspond to successive transformations of the same underlying clinical note.

The *Original* column represents the output after the context generation agent, prior to any anonymization. The next two columns capture intermediate outputs of the anonymization agent: Level 1 (L1), where anonymization is first applied, and Level 2 (L2), where a second LLM-based validation step reviews and refines the anonymized narrative to correct residual privacy leaks. The final column (*Utility*) corresponds to the output of the privacy-utility agent, which enhances clinical detail and readability after anonymization.

Across these stages, we observe a consistent reduction in quasi-identifier exposure from the Original narratives to L1 and further to L2. This confirms that the two-step anonymization process is effective not only for removing direct identifiers but also for progressively reducing quasi-identifier leakage. The improvement from L1 to L2 is particularly notable for age (48.0% to 25.3%), indicating that the validation step plays a key role in addressing residual, harder-to-detect attributes. This demonstrates the importance of iterative refinement in reducing both direct and indirect privacy risks.

In contrast, the Utility stage exhibits a different behavior. While most quasi-identifiers remain at levels comparable to L2, age increases from 25.3% to 44.0%. This reflects the role of the privacy-utility agent, which selectively reintroduces clinically relevant information, such as precise age, to improve interpretability.

To further analyze this trade-off, we next quantify the impact of anonymization and utility refinement on the preservation of clinically relevant information. While the previous analysis focuses on privacy through quasi-identifier exposure, it does not capture how much medical detail is retained across narrative versions. We therefore introduce a set of automated metrics to approximate medical richness and assess whether the reintroduction of selected attributes contributes to improved clinical specificity.

Due to the size and complexity of the MIMIC-III clinical narratives, manual inspection of medical content across all cases and narrative variants is not feasible. Each patient case may contain multiple narrative versions (original, Level 1 anonymized, Level 2 anonymized, and utility-enhanced), resulting in a large volume of text that makes consistent manual comparison difficult.

To address this, we employ automated text-based extraction methods to approximate medical richness across narratives. Specifically, we quantify three aspects of clinical detail: medication mentions, dosage information, and medical term density. Medication mentions are identified using pattern matching based on common pharmaceutical suffixes (e.g., -ine, -ol, -pril, -sartan), which capture typical drug naming conventions in clinical text. Dosage information is extracted using regular expressions that detect numerical values followed by standard medical units (e.g., mg, ml, mcg). Medical term density is estimated using a curated list of common clinical keywords (e.g., diagnosis, treatment, admitted, chronic). These metrics provide an approximate but consistent measure of clinical specificity across different narrative versions, enabling quantitative comparison of how anonymization and utility refinement affect the richness of medical information.

Table 9 shows the results across narrative versions. As shown in Table 9, Level 1 and Level 2 anonymization introduce a modest reduction in clinically actionable detail, reflected by decreases in both medication mentions (40.93 to 38.91) and dosage mentions (7.03 to 6.01). This indicates that while the anonymization process preserves the overall clinical content, certain detailed elements, particularly dosage-related information, are partially removed or generalized.

The utility-enhanced narratives partially recover this loss of detail. In particular, dosage mentions increase from 6.01 in L2 to 6.53 in the utility stage, suggesting that the privacy-utility agent reintroduces clinically actionable information that is important for interpretation. Medication mentions also show a slight recovery (38.91 to 39.37), indicating improved specificity without returning to fully identifiable detail.

Table 9
Medical richness metrics across narrative versions (n=75)

Attribute	Original	Anonymization		Utility
		L1	L2	
Medication mentions (avg count per narrative)	40.93	39.41	38.91	39.37
Dosage mentions (avg count per narrative)	7.03	6.01	6.01	6.53
Medical term density (avg count per narrative)	14.32	14.41	14.28	14.23

Table 10
Examples of medical information restored in utility-enhanced narratives.

Type	L2 Narrative	Utility Narrative
Exact age restoration	30–40 year old female 70-something patient	37-year-old female 80-year-old female patient
Medication specificity	enalapril, lasix	enalapril 15 mg twice daily
Diagnosis refinement	bipolar disorder depression history	bipolar type 2 severe depression with subtype
Clinical detail enrichment	shortness of breath follow-up visit	shortness of breath with edema ICU admission with hypotension and IV fluids

In contrast, medical term density remains largely stable across all narrative versions, demonstrating that the overall semantic clinical content is preserved throughout the pipeline. This suggests that anonymization primarily affects the granularity of information rather than its presence.

Overall, these results confirm that while anonymization reduces fine-grained clinical detail, the utility refinement step effectively restores key elements of clinical specificity, supporting a balanced trade-off between privacy preservation and data usability.

To complement the quantitative analysis, we provide representative examples in Table 10 illustrating how the utility-enhanced narratives restore clinically relevant information. The utility-enhanced narratives introduce incremental but clinically meaningful refinements. These include restoration of exact age values, addition of medication dosages, and more specific diagnostic descriptors. In several cases, broader clinical context is also enriched, such as the inclusion of symptom combinations or acute care interventions. Although these changes are subtle, they significantly enhance clinical interpretability and realism, supporting the value of controlled reintroduction of medical information.

Overall, these results highlight a structured progression across the proposed pipeline, where each agent contributes a distinct and necessary function. The anonymization agent (L1) effectively removes the majority of sensitive attributes, but leaves residual quasi-identifiers. The validation stage (L2) plays a critical role in strengthening privacy guarantees by identifying and correcting remaining leakage that is not fully addressed by initial anonymization alone. Finally, the privacy-utility agent introduces a controlled trade-off by selectively restoring clinically relevant information, improving interpretability without significantly compromising privacy.

These findings demonstrate that no single component is sufficient in isolation. Instead, the combination of context generation, anonymization, validation, and utility refinement enables a balanced and robust solution. This layered design justifies the proposed agentic framework, where privacy preservation and clinical utility are achieved through coordinated, stage-wise refinement rather than a single-pass transformation.

6.10. Computational performance and scalability

To assess the computational feasibility of the proposed framework, we analyze runtime and processing characteristics on the MIMIC-III dataset under the experimental setup described in Section 6.1. The system is executed using a locally deployed Qwen2.5-7B-Instruct model on a single NVIDIA G4 GPU available through Google Colab.

On average, the framework processes a single clinical narrative in 135.4 s (± 65.0). Each narrative is segmented into an average of 3.18 chunks to satisfy LLM context constraints. For each chunk, the pipeline applies multiple stages, including context generation, anonymization, validation, and utility refinement. The observed variation in processing time is primarily driven by differences in narrative length, as longer documents require more chunks and therefore more processing steps. This chunk-based design enables the framework to handle arbitrarily long clinical narratives while maintaining stable execution behavior.

This corresponds to an approximate throughput of 0.44 cases per minute under sequential execution. Importantly, the modular design of the framework allows independent patient cases to be processed in parallel, enabling substantial improvements in throughput in practical deployments. Furthermore, all sensitive processing is performed locally using a 7B parameter model, eliminating the need for external API calls and supporting deployment within hospital infrastructure.

To further assess deployment feasibility, we analyze the financial and latency overhead associated with the public LLM component. Across 100 patient narratives, the average cost of invoking GPT-4o for the final clinical reasoning step is approximately $0.0031 per query, corresponding to $3.1 per 1000 queries. While this introduces additional cost in high-volume settings, it remains manageable in practice given that the public LLM is used only for a single inference step per case.

From a computational perspective, this design adds a single external inference call per case, introducing latency comparable to standard API response times for large language models. Since all preceding stages are executed locally, the framework minimizes both data transfer and external dependency overhead. Additionally, the modular design allows the public LLM component to be replaced with locally hosted models when stricter cost or latency constraints are required.

Overall, these results demonstrate that the proposed framework is computationally feasible and scalable for real-world clinical environments, balancing runtime, cost, and performance while maintaining strong privacy guarantees.

6.11. Discussion

This section examines the key characteristics of the proposed framework in greater detail. It evaluates the system's anonymization performance on clinical data, analyzes its scalability across varying data volumes and complexities, and assesses its potential for generalization to other domains and use cases.

6.11.1. Performance

Our quantitative evaluation presented in Section 6.4 revealed that the proposed anonymization framework performs with a high degree of precision and contextual awareness, particularly when applied to

complex clinical narratives. Across all tested cases, the system consistently avoided anonymizing essential medical terminology (perfect precision), preserving clinical utility while accurately identifying and anonymizing sensitive information. All terms anonymized by the system corresponded to either explicit PII or indirect identifiers, such as family relationships, professions, and incident locations which could pose a reidentification risk if left exposed.

Importantly, the framework also demonstrated the ability to detect and anonymize indirect or contextual cues, which are often overlooked by traditional NER tools. One notable strength is its contextual disambiguation capability, such as accurately interpreting the term "Hunter" when used as both a person's name and a medical syndrome, as discussed in Section 6.6. This level of inference is crucial for sensitive domains like healthcare, where ambiguity can lead to privacy breaches. While the system performed reliably overall, gender references were occasionally missed (as indicated by the few false negatives in Table 3). However, the presence of refinement agents in the pipeline acts as a safeguard where any missed major PII elements are typically caught and corrected in subsequent validation and revision stages.

Beyond performance accuracy, the framework offers significant practical advantages. Its strong contextual reasoning abilities reduce the need for human-in-the-loop verification, thereby lowering operational costs and minimizing potential delays. Furthermore, with carefully crafted prompts and clearly defined agent roles, the system achieves strong performance without requiring domain-specific fine-tuning. This is exemplified by our effective use of a general-purpose, locally hosted LLM, which was able to carry out anonymization tasks across diverse cases without prior exposure to clinical-specific training data. Overall, these results point to a promising and adaptable anonymization solution, with robust context sensitivity and practical benefits for real-world deployment.

6.11.2. Scalability

The scalability of the proposed anonymization framework can be analyzed along two key dimensions: (1) the number of patient records processed, and (2) the size of individual patient records. While the current implementation processes one patient record at a time, the modular agentic design allows future extension to support batch processing. In this envisioned mode, independent components such as data retrieval, narrative generation, and anonymization agents can be parallelized to handle multiple records concurrently, enabling the framework to scale effectively in high-throughput clinical environments.

In terms of record complexity, as the number of visits, diagnoses, medications, and incident reports per patient increases (or in the case of larger, more comprehensive EMRs), the generated narrative length also increases, resulting in more tokens being passed to downstream LLMs. This directly affects the scalability of the framework, particularly with respect to model context limits. For example, the locally hosted Qwen2.5-7B-Instruct model which we adopt in our framework supports a maximum input context of 32,768 tokens, which corresponds to approximately 24,000–26,000 words depending on formatting and language. While most clinical narratives remain within this boundary, highly detailed patient histories may exceed it. To address this, two strategies may be adopted and seamlessly integrated into our modular framework: (1) Switching to larger-context models (e.g., Qwen2.5-72B or GPT-4 Turbo with 128K context), albeit with higher memory and compute requirements, or (2) Using vector stores and chunking narratives into smaller, semantically coherent sections that are processed sequentially and then reassembled. Since LLMs are hosted locally, this does not incur additional API or token-based costs, making chunking a practical and scalable solution.

The only cost associated with processing each case arises when anonymized narratives are sent to public paid LLMs for reasoning or response generation. For instance, OpenAI's GPT-4 Turbo charges approximately $0.01 per 1000 input tokens and $0.03 per 1000 output tokens. For a sample case involving an 8000-token input and a 1000-token output, the total cost would be around $0.11. Hence, chunking larger narratives while performing anonymization locally provides a cost-effective and scalable approach. In summary, the framework is designed to scale with both increasing case volume and narrative complexity. While batch processing is not implemented in the current version, the architecture supports it as a natural extension for future deployments.

6.11.3. Generalizability

An important strength of the proposed anonymization framework lies in its potential for generalization across domains, supported by its modular, agentic design. Anonymization requirements vary significantly between applications, depending on regulatory guidelines, the nature of PII, and the aspects of the data that must be preserved to retain utility.

In the *clinical domain*, for instance, anonymization must retain the chronological structure of events, such as symptom onset, diagnosis dates, and medication timelines. This temporal integrity is crucial for downstream reasoning tasks like clinical decision support or analysis. Consequently, the framework's anonymization agents are designed to recognize and preserve such structures while removing or transforming sensitive identifiers (e.g., names, hospital locations, exact dates). In contrast, in the *financial domain*, where documents such as annual reports or audit summaries require anonymization, strategies that focus on different types of PII, such as client identifiers, account numbers, or specific investment portfolios need to be considered. Moreover, the focus shifts to preserving logical consistency and numerical integrity, ensuring that financial trends and summaries remain interpretable after anonymization. The same can be reflected on the anonymization in the *legal domain*.

Due to the task-specific modularity of our architecture, adapting to such varied contexts requires only minimal modifications. Specifically, the documents that define the set of sensitive entities (PII and quasi-identifiers), the anonymization guidelines, the definition of data utility for the intended domain need to be updated. These modifications can be done without altering the overall pipeline. This makes the framework highly adaptable and generalizable across domains with minimal engineering overhead. Currently, the system supports only English-language inputs. Multilingual support has not yet been integrated and represents an important direction for future work, especially for domains with diverse linguistic and cultural contexts.

7. Conclusion

In this paper, we proposed an end-to-end, contextual integrity-aware agentic LLM system for healthcare applications. The modular architecture leveraged locally hosted LLMs to perform privacy-preserving medical context transformations then utilized public LLMs for generating treatment plans and medical suggestions without exposing sensitive data. This design addressed the persistent privacy-utility tradeoff that often hinders the use of clinical text in downstream applications. Our system consists of specialized agents coordinated by an orchestrator that enables a structured and privacy-conscious workflow. This workflow guarantees that sensitive data is handled exclusively by the agents responsible for its processing, while preventing unnecessary exposure to agents that do not need access. Through iterative validation and refinement, the system ensures that only anonymized narratives with high clinical utility are submitted to public LLMs. This layered approach provides strong privacy guarantees while preserving clinical relevance.

Our comparative evaluation against five state-of-the-art NER tools demonstrated that general-purpose and biomedical models often suffer from low precision, leading to over-anonymization and loss of critical context. In contrast, our method achieved high precision and strong recall, as evidenced by both qualitative and quantitative analyses.

Additional evaluation on real-world clinical notes from the MIMIC-III dataset demonstrated strong generalization to unstructured clinical narratives. Beyond anonymization performance, we evaluated medical richness, showing that anonymized outputs preserve clinically relevant information and maintain high semantic fidelity to the original records. These results highlight the effectiveness of our framework in jointly optimizing privacy protection, re-identification robustness, and clinical utility.

Future work will focus on extending the framework to support multi-modal data and multilingual inputs, further enhancing its applicability in real-world, diverse clinical environments.

CRediT authorship contribution statement

Rana Azzam: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Formal analysis, Conceptualization. **Ahmad Musamih:** Writing – review & editing, Conceptualization. **Senay Gebreab:** Writing – review & editing, Conceptualization. **Khaled Salah:** Writing – review & editing, Visualization, Supervision, Project administration, Funding acquisition, Conceptualization. **Mohammed Omar:** Supervision, Project administration, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This research was funded by the Socio-Technical Systems Lab (STSL), Khalifa University of Science and Technology (KU-STSL). During the preparation and review of this work, the authors used ChatGPT to improve readability and language. After using this tool, the authors reviewed and edited the content as needed and took full responsibility for the content of the publication.

Code and data availability

The synthetic dataset and implementation of the proposed agentic LLM framework are available from the authors upon request.

References

[1] L. Li, J. Zhou, Z. Gao, W. Hua, L. Fan, H. Yu, L. Hagen, Y. Zhang, T.L. Assimes, L. Hemphill, S. Ma, A scoping review of using large language models (LLMs) to investigate electronic health records (EHRs), 2024, arXiv:2405.03066, URL https://arxiv.org/abs/2405.03066.

[2] C.-H. Lin, C.-F. Kuo, Roles and potential of large language models in healthcare: A comprehensive review, Biomed. J. (2025) 100868, http://dx.doi.org/10.1016/j.bj.2025.100868, Epub ahead of print.

[3] D. Li, N. Jiang, K. Huang, R. Tu, S. Ouyang, H. Yu, L. Qiao, C. Yu, T. Zhou, D. Tong, Q. Wang, M. Li, X. Zeng, Y. Tian, X. Tian, J. Li, From questions to clinical recommendations: Large language models driving evidence-based clinical decision making, 2025, arXiv:2505.10282, URL https://arxiv.org/abs/2505.10282.

[4] J.C.L. Ong, L. Jin, K. Elangovan, G.Y.S. Lim, D.Y.Z. Lim, G.G.R. Sng, Y. Ke, J.Y.M. Tung, R.J. Zhong, C.M.Y. Koh, K.Z.H. Lee, X. Chen, J.K. Chng, A. Than, K.J. Goh, D.S.W. Ting, Development and testing of a novel large language model-based clinical decision support systems for medication safety in 12 clinical specialties, 2024, arXiv:2402.01741, URL https://arxiv.org/abs/2402.01741.

[5] S. Alder, Healthcare data breach statistics, 2024, https://www.hipaajournal.com/healthcare-data-breach-statistics/. (Accessed: 07 July 2025).

[6] Health insurance portability and accountability act of 1996 (HIPAA), 1996, Public Law 104-191. URL https://aspe.hhs.gov/reports/health-insurance-portability-accountability-act-1996.

[7] General data protection regulation (GDPR), 2016, URL https://gdpr.eu/article-4-definitions.

[8] A. Sepas, A.H. Bangash, O. Alraoui, K. El Emam, A. El-Hussuna, Algorithms to anonymize structured medical and healthcare data: A systematic review, Front. Bioinform. 2 (2022) 984807, http://dx.doi.org/10.3389/fbinf.2022.984807.

[9] O. Yermilov, V. Raheja, A. Chernodub, Privacy- and utility-preserving NLP with anonymized data: A case study of pseudonymization, 2023, arXiv:2306.05561, URL https://arxiv.org/abs/2306.05561.

[10] A. Dorca Josa, M. Bleda-Bejar, Local LLMS: Safeguarding data privacy in the age of generative AI. A case study at the university of andorra, in: ICERI2024 Proceedings, in: 17th annual International Conference of Education, Research and Innovation, IATED, 2024, pp. 7879–7888, http://dx.doi.org/10.21125/iceri.2024.1931.

[11] S. Rahman, L.Y. Jiang, S. Gabriel, Y. Aphinyanaphongs, E.K. Oermann, R. Chunara, Generalization in healthcare AI: Evaluation of a clinical large language model, 2024, arXiv:2402.10965, URL https://arxiv.org/abs/2402.10965.

[12] T. Masterman, S. Besen, M. Sawtell, A. Chao, The landscape of emerging AI agent architectures for reasoning, planning, and tool calling: A survey, 2024, arXiv:2404.11584, URL https://arxiv.org/abs/2404.11584.

[13] T. Meurers, K. Otte, H. Abu Attieh, F. Briki, J. Despraz, M. Halilovic, B. Kaabachi, V. Milicevic, A. Müller, G. Papapostolou, F.N. Wirth, J.L. Raisaro, F. Prasser, A quantitative analysis of the use of anonymization in biomedical research, NPJ Digit. Med. 8 (1) (2025) 279, http://dx.doi.org/10.1038/s41746-025-01644-9.

[14] L. Sweeney, k-anonymity: a model for protecting privacy, Int. J. Uncertain. Fuzziness Knowl.-Based Syst. (ISSN: 0218-4885) 10 (5) (2002) 557–570, http://dx.doi.org/10.1142/S0218488502001648.

[15] A. Machanavajjhala, J. Gehrke, D. Kifer, M. Venkitasubramaniam, L-diversity: privacy beyond k-anonymity, in: 22nd International Conference on Data Engineering, ICDE'06, 2006, http://dx.doi.org/10.1109/ICDE.2006.1, 24–24.

[16] N. Li, T. Li, S. Venkatasubramanian, T-closeness: Privacy beyond k-anonymity and l-diversity, in: 2007 IEEE 23rd International Conference on Data Engineering, 2007, pp. 106–115, http://dx.doi.org/10.1109/ICDE.2007.367856.

[17] C. Dwork, F. McSherry, K. Nissim, A. Smith, Calibrating noise to sensitivity in private data analysis, in: S. Halevi, T. Rabin (Eds.), Theory of Cryptography, Springer Berlin Heidelberg, Berlin, Heidelberg, ISBN: 978-3-540-32732-5, 2006, pp. 265–284.

[18] Y. Rubner, C. Tomasi, L.J. Guibas, The earth mover's distance as a metric for image retrieval, Int. J. Comput. Vis. 40 (2) (2000) 99–121, http://dx.doi.org/10.1023/A:1026543900054, URL https://doi.org/10.1023/A:1026543900054.

[19] I. Baroud, L. Raithel, S. Möller, R. Roller, Beyond de-identification: A structured approach for defining and detecting indirect identifiers in medical texts, 2025, http://dx.doi.org/10.48550/arXiv.2502.13342, arXiv preprint arXiv:2502.13342, arXiv:2502.13342.

[20] I. Neamatullah, M.M. Douglass, L. wei H. Lehman, A. Reisner, M. Villarroel, W.J. Long, P. Szolovits, G.B. Moody, R.G. Mark, G.D. Clifford, Automated de-identification of free-text medical records, BMC Med. Inform. Decis. Mak. 8 (1) (2008) 32, http://dx.doi.org/10.1186/1472-6947-8-32.

[21] M.E. Peters, M. Neumann, M. Iyyer, M. Gardner, C. Clark, K. Lee, L. Zettlemoyer, Deep contextualized word representations, 2018, CoRR abs/1802.05365, arXiv:1802.05365, URL http://arxiv.org/abs/1802.05365.

[22] D. Asimopoulos, I. Siniosoglou, V. Argyriou, T. Karamitsou, E. Fountoukidis, S.K. Goudos, I.D. Moscholios, K.E. Psannis, P. Sarigiannidis, Benchmarking advanced text anonymisation methods: A comparative study on novel and traditional approaches, 2024, arXiv preprint arXiv:2404.14465, URL https://arxiv.org/abs/2404.14465. (Accessed: 04 June 2025).

[23] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A.N. Gomez, L. Kaiser, I. Polosukhin, Attention is all you need, 2023, arXiv:1706.03762, URL https://arxiv.org/abs/1706.03762.

[24] J. Devlin, M. Chang, K. Lee, K. Toutanova, BERT: pre-training of deep bidirectional transformers for language understanding, 2018, CoRR abs/1810.04805, arXiv:1810.04805, URL http://arxiv.org/abs/1810.04805.

[25] J. Lee, W. Yoon, S. Kim, D. Kim, S. Kim, C.H. So, J. Kang, Biobert: a pre-trained biomedical language representation model for biomedical text mining, Bioinformatics 36 (4) (2019) 1234–1240, http://dx.doi.org/10.1093/bioinformatics/btz682.

[26] E. Alsentzer, J.R. Murphy, W. Boag, W. Weng, D. Jin, T. Naumann, M.B.A. McDermott, Publicly available clinical BERT embeddings, 2019, CoRR abs/1904.03323, arXiv:1904.03323, URL http://arxiv.org/abs/1904.03323.

[27] J.C.L. Ong, L. Jin, K. Elangovan, G.Y.S. Lim, D.Y.Z. Lim, G.G.R. Sng, Y. Ke, J.Y.M. Tung, R.J. Zhong, C.M.Y. Koh, K.Z.H. Lee, X. Chen, J.K. Chng, A. Than, K.J. Goh, D.S.W. Ting, Development and testing of a novel large language model-based clinical decision support systems for medication safety in 12 clinical specialties, 2024, arXiv:2402.01741, URL https://arxiv.org/abs/2402.01741.

[28] D. Oniani, X. Wu, S. Visweswaran, S. Kapoor, S. Kooragayalu, K. Polanska, Y. Wang, Enhancing large language models for clinical decision support by incorporating clinical practice guidelines, 2024, arXiv:2401.11120, URL https://arxiv.org/abs/2401.11120.

[29] R. Poulain, H. Fayyaz, R. Beheshti, Bias patterns in the application of LLMs for clinical decision support: A comprehensive study, 2024, arXiv:2404.15149, URL https://arxiv.org/abs/2404.15149.

[30] M. Lyu, C. Peng, X. Li, P. Balian, J. Bian, Y. Wu, Automatic summarization of doctor-patient encounter dialogues using large language model through prompt tuning, 2024, arXiv:2403.13089, URL https://arxiv.org/abs/2403.13089.

[31] H. Wang, C. Gao, B. Liu, Q. Xu, G. Hussein, M.E. Labban, K. Iheasirim, H. Korsapati, C. Outcalt, J. Sun, Towards adapting open-source large language models for expert-level clinical note generation, 2025, arXiv:2405.00715, URL https://arxiv.org/abs/2405.00715.

[32] R. Srivastava, S. Prasad, L. Bhat, S. Deshpande, B. Das, K. Jadhav, MedPromptExtract (medical data extraction tool): Anonymization and hi-fidelity automated data extraction using NLP and prompt engineering, 2024, arXiv:2405.02664, URL https://arxiv.org/abs/2405.02664.

[33] Y. Labrak, A. Bazoge, E. Morin, P.-A. Gourraud, M. Rouvier, R. Dufour, BioMistral: A collection of open-source pretrained large language models for medical domains, 2024, arXiv:2402.10373, URL https://arxiv.org/abs/2402.10373.

[34] R. Luo, L. Sun, Y. Xia, T. Qin, S. Zhang, H. Poon, T.-Y. Liu, BioGPT: generative pre-trained transformer for biomedical text generation and mining, Brief. Bioinform. 23 (6) (2022) http://dx.doi.org/10.1093/bib/bbac409.

[35] Z. Liu, Y. Huang, X. Yu, L. Zhang, Z. Wu, C. Cao, H. Dai, L. Zhao, Y. Li, P. Shu, F. Zeng, L. Sun, W. Liu, D. Shen, Q. Li, T. Liu, D. Zhu, X. Li, DeID-GPT: Zero-shot medical text de-identification by GPT-4, 2023, arXiv:2303.11032, URL https://arxiv.org/abs/2303.11032.

[36] E. Bagdasarian, R. Yi, S. Ghalebikesabi, P. Kairouz, M. Gruteser, S. Oh, B. Balle, D. Ramage, AirGapAgent: Protecting privacy-conscious conversational agents, 2024, arXiv:2405.05175, URL https://arxiv.org/abs/2405.05175.

[37] I.C. Wiest, M.-E. Leßmann, F. Wolf, D. Ferber, M. Van Treeck, J. Zhu, M.P. Ebert, C.B. Westphalen, M. Wermke, J.N. Kather, Anonymizing medical documents with local, privacy preserving large language models: The LLM-anonymizer, 2024, http://dx.doi.org/10.1101/2024.06.11.24308355, MedRxiv, arXiv:https://www.medrxiv.org/content/early/2024/06/13/2024.06.11.24308355.full.pdf, URL https://www.medrxiv.org/content/early/2024/06/13/2024.06.11.24308355.

[38] D. Pissarra, I. Curioso, J. Alveira, D. Pereira, B. Ribeiro, T. Souper, V. Gomes, A.V. Carreiro, V. Rolla, Unlocking the potential of large language models for clinical text anonymization: A comparative study, 2024, arXiv:2406.00062, URL https://arxiv.org/abs/2406.00062.

[39] T. Yang, X. Zhu, I. Gurevych, Robust utility-preserving text anonymization based on large language models, 2024, arXiv:2407.11770, URL https://arxiv.org/abs/2407.11770.

[40] H. Chase, LangChain, 2022, https://github.com/langchain-ai/langchain. (Accessed: 17 June 2025).

[41] M. Honnibal, I. Montani, S. Van Landeghem, A. Boyd, spaCy: Industrial-strength natural language processing in Python, 2020, https://Spacy.Io.

[42] P. Qi, Y. Zhang, Y. Zhang, J. Bolton, C.D. Manning, Stanza: A python natural language processing toolkit for many human languages, in: A. Celikyilmaz, T.-H. Wen (Eds.), Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics: System Demonstrations, Association for Computational Linguistics, Online, 2020, pp. 101–108, http://dx.doi.org/10.18653/v1/2020.acl-demos.14, URL https://aclanthology.org/2020.acl-demos.14/.

[43] Ö. Uzuner, Y. Luo, P. Szolovits, Evaluating the state-of-the-art in automatic de-identification, J. Am. Med. Informatics Assoc. 14 (5) (2007) 550–563, http://dx.doi.org/10.1197/jamia.M2444.

[44] A. Akbik, D. Blythe, R. Vollgraf, Contextual string embeddings for sequence labeling, in: E.M. Bender, L. Derczynski, P. Isabelle (Eds.), Proceedings of the 27th International Conference on Computational Linguistics, Association for Computational Linguistics, Santa Fe, New Mexico, USA, 2018, pp. 1638–1649, URL https://aclanthology.org/C18-1139/.

[45] T. Wolf, L. Debut, V. Sanh, J. Chaumond, C. Delangue, A. Moi, P. Cistac, T. Rault, R. Louf, M. Funtowicz, J. Davison, S. Shleifer, P. von Platen, C. Ma, Y. Jernite, J. Plu, C. Xu, T. Le Scao, S. Gugger, M. Drame, Q. Lhoest, A. Rush, Transformers: State-of-the-art natural language processing, in: Q. Liu, D. Schlangen (Eds.), Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing: System Demonstrations, Association for Computational Linguistics, Online, 2020, pp. 38–45, http://dx.doi.org/10.18653/v1/2020.emnlp-demos.6, URL https://aclanthology.org/2020.emnlp-demos.6/.

[46] E.F. Tjong Kim Sang, F. De Meulder, Introduction to the coNLL-2003 shared task: Language-independent named entity recognition, in: Proceedings of the Seventh Conference on Natural Language Learning At HLT-NAACL 2003, 2003, pp. 142–147, URL https://aclanthology.org/W03-0419/.

[47] E. Shlezinger, et al., Presidio: Data protection and de-identification SDK, 2021, https://Microsoft.Github.Io/Presidio/.

[48] A. Johnson, T. Pollard, R. Mark, MIMIC-III Clinical Database, PhysioNet (2016) http://dx.doi.org/10.13026/C2XW26, Version 1.4.

[49] A.E.W. Johnson, T.J. Pollard, L. Shen, H.L. Li-Wei, M. Feng, M. Ghassemi, B. Moody, P. Szolovits, L.A. Celi, R.G. Mark, MIMIC-III, a freely accessible critical care database, Sci. Data 3 (2016) 160035, http://dx.doi.org/10.1038/sdata.2016.35.

[50] A.L. Goldberger, L.A.N. Amaral, L. Glass, J.M. Hausdorff, P.C. Ivanov, R.G. Mark, J.E. Mietus, G.B. Moody, C.-K. Peng, H.E. Stanley, PhysioBank, PhysioToolkit, and PhysioNet: Components of a new research resource for complex physiologic signals, Circulation 101 (23) (2000) e215–e220, http://dx.doi.org/10.1161/01.CIR.101.23.e215, RRID:SCR_007345.